# Functional Analysis

Exercise

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# Functional Analysis: Introduction to Further Topics in Analysis

## Chapter 1. $L^p$ Spaces and Banach Spaces

**1.** Consider $L^p = L^p(\mathbb{R}^d)$ with Lebesgue measure. Let $f_0(x) = |x|^{-\alpha}$ if $|x| < 1$, $f_0(x) = 0$ for $|x| \geq 1$; also let $f_\infty(x) = |x|^{-\alpha}$ if $|x| \geq 1$, $f_\infty(x) = 0$ when $|x| < 1$. Show that:

(a) $f_0 \in L^p$ if and only if $p\alpha < d$.

(b) $f_\infty \in L^p$ if and only if $d < p\alpha$.

(c) What happens if in the definitions of $f_0$ and $f_\infty$ we replace $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2/|x|))$ for $|x| < 1$, and $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2|x|))$ for $|x| \geq 1$?

**2.** Consider the spaces $L^p(\mathbb{R}^d)$, when $0 < p < \infty$.

(a) Show that if $\|f+g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$ for all $f$ and $g$, then necessarily $p \geq 1$.

(b) Consider $L^p(\mathbb{R})$ where $0 < p < 1$. Show that there are no bounded linear functionals on this space. In other words, if $\ell$ is a linear function $L^p(\mathbb{R}) \to \mathbb{C}$ that satisfies

$$|\ell(f)| \leq M\|f\|_{L^p(\mathbb{R})} \qquad \text{for all } f \in L^p(\mathbb{R}) \text{ and some } M > 0,$$

then $\ell = 0$.

[Hint: For (a), prove that if $0 < p < 1$ and $x, y > 0$, then $x^p + y^p > (x+y)^p$. For (b), let $F$ be defined by $F(x) = \ell(\chi_x)$, where $\chi_x$ is the characteristic function of $[0, x]$, and consider $F(x) - F(y)$.]

**3.** If $f \in L^p$ and $g \in L^q$, both not identically equal to zero, show that equality holds in Hölder's inequality (Theorem 1.1) if and only if there exist two non-zero constants $a, b \geq 0$ such that $a|f(x)|^p = b|g(x)|^q$ for a.e. $x$.

**4.** Suppose $X$ is a measure space and $0 < p < 1$.

(a) Prove that $\|fg\|_{L^1} \geq \|f\|_{L^p}\|g\|_{L^q}$. Note that $q$, the conjugate exponent of $p$, is negative.

(b) Suppose $f_1$ and $f_2$ are non-negative. Then $\|f_1 + f_2\|_{L^p} \geq \|f_1\|_{L^p} + \|f_2\|_{L^p}$.

(c) The function $d(f, g) = \|f - g\|_{L^p}^p$ for $f, g \in L^p$ defines a metric on $L^p(X)$.

**5.** Let $X$ be a measure space. Using the argument to prove the completeness of $L^p(X)$, show that if the sequence $\{f_n\}$ converges to $f$ in the $L^p$ norm, then a subsequence of $\{f_n\}$ converges to $f$ almost everywhere.

**6.** Let $(X, \mathcal{F}, \mu)$ be a measure space. Show that:

(a) The simple functions are dense in $L^\infty(X)$ if $\mu(X) < \infty$, and;

(b) The simple functions are dense in $L^p(X)$ for $1 \leq p < \infty$.

[Hint: For (a), use $E_{\ell,j} = \{x \in X : \frac{M\ell}{j} \leq f(x) < \frac{M(\ell+1)}{j}\}$ where $-j \leq \ell \leq j$, and $M = \|f\|_{L^\infty}$. Then consider the functions $f_j$ that equal $M\ell/j$ on $E_{\ell,j}$. For (b) use a construction similar to that in (a).]

**7.** Consider the $L^p$ spaces, $1 \leq p < \infty$, on $\mathbb{R}^d$ with Lebesgue measure. Prove that:

(a) The family of continuous functions with compact support is dense in $L^p$, and in fact:

(b) The family of indefinitely differentiable functions with compact support is dense in $L^p$.

The cases of $L^1$ and $L^2$ are in Theorem 2.4, Chapter 2 of Book III, and Lemma 3.1, Chapter 5 of Book III.

**8.** Suppose $1 \leq p < \infty$, and that $\mathbb{R}^d$ is equipped with Lebesgue measure. Show that if $f \in L^p(\mathbb{R}^d)$, then

$$\|f(x+h) - f(x)\|_{L^p} \to 0 \qquad \text{as } |h| \to 0.$$

Prove that this fails when $p = \infty$.

[Hint: By the previous exercise, the continuous functions with compact support are dense in $L^p(\mathbb{R}^d)$ for $1 \leq p < \infty$. See also Theorem 2.4 and Proposition 2.5 in Chapter 2 of Book III.]

**9.** Suppose $X$ is a measure space and $1 \leq p_0 < p_1 \leq \infty$.

(a) Consider $L^{p_0} \cap L^{p_1}$ equipped with

$$\|f\|_{L^{p_0} \cap L^{p_1}} = \|f\|_{L^{p_0}} + \|f\|_{L^{p_1}}.$$

Show that $\|\cdot\|_{L^{p_0} \cap L^{p_1}}$ is a norm, and that $L^{p_0} \cap L^{p_1}$ (with this norm) is a Banach space.

(b) Suppose $L^{p_0} + L^{p_1}$ is defined as the vector space of measurable functions $f$ on $X$ that can be written as a sum $f = f_0 + f_1$ with $f_0 \in L^{p_0}$ and $f_1 \in L^{p_1}$. Consider

$$\|f\|_{L^{p_0} + L^{p_1}} = \inf \left\{ \|f_0\|_{L^{p_0}} + \|f_1\|_{L^{p_1}} \right\},$$

where the infimum is taken over all decompositions $f = f_0 + f_1$ with $f_0 \in L^{p_0}$ and $f_1 \in L^{p_1}$. Show that $\|\cdot\|_{L^{p_0} + L^{p_1}}$ is a norm, and that $L^{p_0} + L^{p_1}$ (with this norm) is a Banach space.

(c) Show that $L^p \subset L^{p_0} + L^{p_1}$ if $p_0 \leq p \leq p_1$.

**10.** A measure space $(X, \mu)$ is **separable** if there is a countable family of measurable subsets $\{E_k\}_{k=1}^{\infty}$ so that if $E$ is any measurable set of finite measure, then

$$\mu(E \triangle E_{n_k}) \to 0 \qquad \text{as } k \to 0$$

for an appropriate subsequence $\{n_k\}$ which depends on $E$. Here $A \triangle B$ denotes the symmetric difference of the sets $A$ and $B$, that is,

$$A \triangle B = (A - B) \cup (B - A).$$

(a) Verify that $\mathbb{R}^d$ with the usual Lebesgue measure is separable.

(b) The space $L^p(X)$ is **separable** if there exists a countable collection of elements $\{f_n\}_{n=1}^{\infty}$ in $L^p$ that is dense. Prove that if the measure space $X$ is separable, then $L^p$ is separable when $1 \leq p < \infty$.

**11.** In light of the previous exercise, prove the following:

(a) Show that the space $L^\infty(\mathbb{R})$ is not separable by constructing for each $a \in \mathbb{R}$ an $f_a \in L^\infty$, with $\|f_a - f_b\| \geq 1$, if $a \neq b$.

(b) Do the same for the dual space of $L^\infty(\mathbb{R})$.

**12.** Suppose the measure space $(X, \mu)$ is separable as defined in Exercise 10. Let $1 \leq p < \infty$ and $1/p + 1/q = 1$. A sequence $\{f_n\}$ with $f_n \in L^p$ is said to converge to $f \in L^p$ **weakly** if

$$\int f_n g \, d\mu \to \int f g \, d\mu \qquad \text{for every } g \in L^q. \tag{22}$$

(a) Verify that if $\|f - f_n\|_{L^p} \to 0$, then $f_n$ converges to $f$ weakly.

(b) Suppose $\sup_n \|f_n\|_{L^p} < \infty$. Then, to verify weak convergence it suffices to check (22) for a dense subset of functions $g$ in $L^q$.

(c) Suppose $1 < p < \infty$. Show that if $\sup_n \|f_n\|_{L^p} < \infty$, then there exists $f \in L^p$, and a subsequence $\{n_k\}$ so that $f_{n_k}$ converges weakly to $f$.

Part (c) is known as the "weak compactness" of $L^p$ for $1 < p < \infty$, which fails when $p = 1$ as is seen in the exercise below.

[Hint: For (b) use Exercise 10 (b).]

**13.** Below are some examples illustrating weak convergence.

(a) $f_n(x) = \sin(2\pi nx)$ in $L^p([0,1])$. Show that $f_n \to 0$ weakly.

(b) $f_n(x) = n^{1/p}\chi(nx)$ in $L^p(\mathbb{R})$. Then $f_n \to 0$ weakly if $p > 1$, but not when $p = 1$. Here $\chi$ denotes the characteristic function of $[0,1]$.

(c) $f_n(x) = 1 + \sin(2\pi nx)$ in $L^1([0,1])$. Then $f_n \to 1$ weakly also in $L^1([0,1])$, $\|f_n\|_{L^1} = 1$, but $\|f_n - 1\|_{L^1}$ does not converge to zero. Compare with Problem 6 part (d).

**14.** Suppose $X$ is a measure space, $1 < p < \infty$, and suppose $\{f_n\}$ is a sequence of functions with $\|f_n\|_{L^p} \leq M < \infty$.

(a) Prove that if $f_n \to f$ a.e. then $f_n \to f$ weakly.

(b) Show that the above result may fail if $p = 1$.

(c) Show that if $f_n \to f_1$ a.e. and $f_n \to f_2$ weakly, then $f_1 = f_2$ a.e.

**15. Minkowski's inequality for integrals.** Suppose $(X_1, \mu_1)$ and $(X_2, \mu_2)$ are two measure spaces, and $1 \leq p \leq \infty$. Show that if $f(x_1, x_2)$ is measurable on $X_1 \times X_2$ and non-negative, then

$$\left\| \int f(x_1, x_2)\, d\mu_2 \right\|_{L^p(X_1)} \leq \int \|f(x_1, x_2)\|_{L^p(X_1)}\, d\mu_2.$$

Extend this statement to the case when $f$ is complex-valued and the right-hand side of the inequality is finite.

[Hint: For $1 < p < \infty$, use a combination of Hölder's inequality, and its converse in Lemma 4.2.]

**16.** Prove that if $f_j \in L^{p_j}(X)$, where $X$ is a measure space, $j = 1, \ldots, N$, and $\sum_{j=1}^{N} 1/p_j = 1$ with $p_j \geq 1$, then

$$\| \prod_{j=1}^{N} f_j \|_{L^1} \leq \prod_{j=1}^{N} \|f\|_{L^{p_j}} .$$

This is the multiple Hölder inequality.

**17.** The **convolution** of $f$ and $g$ on $\mathbb{R}^d$ equipped with the Lebesgue measure is defined by

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\,dy.$$

(a) If $f \in L^p$, $1 \leq p \leq \infty$, and $g \in L^1$, then show that for almost every $x$ the integrand $f(x-y)g(y)$ is integrable in $y$, hence $f * g$ is well defined. Moreover, $f * g \in L^p$ with

$$\|f * g\|_{L^p} \leq \|f\|_{L^p}\|g\|_{L^1}.$$

(b) A version of (a) applies when $g$ is replaced by a finite Borel measure $\mu$: if $f \in L^p$, with $1 \leq p \leq \infty$, define

$$(f * \mu)(x) = \int_{\mathbb{R}^d} f(x-y)\,d\mu(y),$$

and show that $\|f * \mu\|_{L^p} \leq \|f\|_{L^p}|\mu|(\mathbb{R}^d)$.

(c) Prove that if $f \in L^p$ and $g \in L^q$, where $p$ and $q$ are conjugate exponents, then $f * g \in L^\infty$ with $\|f * g\|_{L^\infty} \leq \|f\|_{L^p}\|g\|_{L^q}$. Moreover, the convolution $f * g$ is uniformly continuous on $\mathbb{R}$, and if $1 < p < \infty$, then $\lim_{|x|\to\infty}(f * g)(x) = 0$.

[Hint: For (a) and (b) use the Minkowski inequality for integrals in Exercise 15. For part (c), use Exercise 8.]

**18.** We consider the $L^p$ spaces with **mixed norm**, in a special case that is useful is several contexts.

We take as our underlying space the product space $\{(x,t)\} = \mathbb{R}^d \times \mathbb{R}$, with the product measure $dx\,dt$, where $dx$ and $dt$ are Lebesgue measures on $\mathbb{R}^d$ and $\mathbb{R}$ respectively. We define $L^r_t(L^p_x) = L^{p,r}$, with $1 \le p \le \infty$, $1 \le r \le \infty$, to be the space of equivalence classes of jointly measurable functions $f(x,t)$ for which the norm

$$\|f\|_{L^{p,r}} = \left( \int_{\mathbb{R}} \left( \int_{\mathbb{R}^d} |f(x,t)|^p \, dx \right)^{\frac{r}{p}} \, dt \right)^{\frac{1}{r}}$$

is finite (when $p < \infty$ and $r < \infty$), and an obvious variant when $p = \infty$ or $r = \infty$.

(a) Verify that $L^{p,r}$ with this norm is complete, and hence is a Banach space.

(b) Prove the general form of Hölder's inequality in this context

$$\int_{\mathbb{R}^d \times \mathbb{R}} |f(x,t)g(x,t)| \, dx\, dt \le \|f\|_{L^{p,r}} \|g\|_{L^{p',r'}},$$

with $1/p + 1/p' = 1$ and $1/r + 1/r' = 1$.

(c) Show that if $f$ is integrable over all sets of finite measure, then

$$\|f\|_{L^{p,r}} = \sup \left| \int_{\mathbb{R}^d \times \mathbb{R}} f(x,t)g(x,t) \, dxdt \right|,$$

with the sup taken over all $g$ that are simple and $\|g\|_{L^{p',r'}} \le 1$.

(d) Conclude that the dual space of $L^{p,r}$ is $L^{p',r'}$, if $1 \le p < \infty$, and $1 \le r < \infty$.

**19. Young's inequality.** Suppose $1 \le p, q, r \le \infty$. Prove the following on $\mathbb{R}^d$:

$$\|f * g\|_{L^q} \le \|f\|_{L^p} \|g\|_{L^r} \qquad \text{whenever } 1/q = 1/p + 1/r - 1.$$

Here, $f * g$ denotes the convolution of $f$ and $g$ as defined in Exercise 17.
[Hint: Assume $f, g \ge 0$, and use the decomposition

$$f(y)g(x-y) = f(y)^a g(x-y)^b [f(y)^{1-a} g(x-y)^{1-b}]$$

for appropriate $a$ and $b$, together with Exercise 16 to find that

$$\left| \int f(y)g(x-y)\, dy \right| \le \|f\|_{L^p}^{1-q/p} \|g\|_{L^q}^{1-q/r} \left( \int |f(y)|^p |g(x-y)|^r \, dy \right)^{\frac{1}{q}}.]$$

**20.** Suppose $X$ is a measure space, $0 < p_0 < p < p_1 \leq \infty$, and $f \in L^{p_0}(X) \cap L^{p_1}(X)$. Then $f \in L^p(X)$ and

$$\|f\|_{L^p} \leq \|f\|_{L^{p_0}}^{1-t} \|f\|_{L^{p_1}}^{t}, \qquad \text{if } t \text{ is chosen so that } \tfrac{1}{p} = \tfrac{1-t}{p_0} + \tfrac{t}{p_1}.$$

**21.** Recall the definition of a convex function. (See Problem 4, Chapter 3, in Book III.) Suppose $\varphi$ is a non-negative convex function on $\mathbb{R}$ and $f$ is real-valued and integrable on a measure space $X$, with $\mu(X) = 1$. Then we have **Jensen's inequality**:

$$\varphi\left(\int_X f\,d\mu\right) \leq \int_X \varphi(f)\,d\mu.$$

Note that if $\varphi(t) = |t|^p$, $1 \leq p$, then $\varphi$ is convex and the above can be obtained from Hölder's inequality. Another interesting case is $\varphi(t) = e^{at}$.

[Hint: Since $\varphi$ is convex, one has, $\varphi(\sum_{j=1}^N a_j x_j) \leq \sum_{j=1}^N a_j \varphi(x_j)$, whenever $a_j, x_j$ are real, $a_j \geq 0$, and $\sum_{j=1}^N a_j = 1$.]

**22. Another inequality of Young**. Suppose $\varphi$ and $\psi$ are both continuous, *strictly* increasing functions on $[0, \infty)$ that are inverses of each other, that is, $(\varphi \circ \psi)(x) = x$ for all $x \geq 0$. Let

$$\Phi(x) = \int_0^x \varphi(u)\, du \qquad \text{and} \qquad \Psi(x) = \int_0^x \psi(u)\, du.$$

(a) Prove: $ab \leq \Phi(a) + \Psi(b)$ for all $a, b \geq 0$.

In particular, if $\varphi(x) = x^{p-1}$ and $\psi(y) = y^{q-1}$ with $1 < p < \infty$ and $1/p + 1/q = 1$, then we get $\Phi(x) = x^p/p$, $\Psi(y) = y^q/q$, and

$$A^\theta B^{1-\theta} \leq \theta A + (1-\theta)B \qquad \text{for all } A, B \geq 0 \text{ and } 0 \leq \theta \leq 1.$$

(b) Prove that we have equality in Young's inequality only if $b = \varphi(a)$ (that is, $a = \psi(b)$).

[Hint: Consider the area $ab$ of the rectangle whose vertices are $(0,0)$, $(a,0)$, $(0,b)$ and $(a,b)$, and compare it to areas "under" the curves $y = \Phi(x)$ and $x = \Psi(y)$.]

**23.** Let $(X, \mu)$ be a measure space and suppose $\Phi(t)$ is a continuous, convex, and increasing function on $[0, \infty)$, with $\Phi(0) = 0$. Define

$$L^\Phi = \{f \text{ measurable} : \int_X \Phi(|f(x)|/M)\, d\mu < \infty \text{ for some } M > 0\},$$

and

$$\|f\|_\Phi = \inf_{M>0} \int_X \Phi(|f(x)|/M)\, d\mu \leq 1.$$

Prove that:

(a) $L^\Phi$ is a vector space.

(b) $\|\cdot\|_{L^\Phi}$ is a norm.

(c) $L^\Phi$ is complete in this norm.

The Banach spaces $L^\Phi$ are called **Orlicz spaces**. Note that in the special case $\Phi(t) = t^p$, $1 \leq p < \infty$, then $L^\Phi = L^p$.

[Hint: Observe that if $f \in L^\Phi$, then $\lim_{N\to\infty} \int_X \Phi(|f|/N)\, d\mu = 0$. Also, use the fact that there exists $A > 0$ so that $\Phi(t) \geq At$ for all $t \geq 0$.]

**24.** Let $1 \le p_0 < p_1 < \infty$.

(a) Consider the Banach space $L^{p_0} \cap L^{p_1}$ with norm $\|f\|_{L^{p_0}\cap L^{p_1}} = \|f\|_{L^{p_0}} + \|f\|_{L^{p_1}}$. (See Exercise 9.) Let

$$\Phi(t) = \begin{cases} t^{p_0} & \text{if } 0 \le t \le 1, \\ t^{p_1} & \text{if } 1 \le t < \infty. \end{cases}$$

Show that $L^\Phi$ with its norm is equivalent to the space $L^{p_0} \cap L^{p_1}$. In other words, there exist $A, B > 0$, so that

$$A\|f\|_{L^{p_0}\cap L^{p_1}} \le \|f\|_{L^\Phi} \le B\|f\|_{L^{p_0}\cap L^{p_1}}.$$

(b) Similarly, consider the Banach space $L^{p_0} + L^{p_1}$ with its norm as defined in Exercise 9. Let

$$\Psi(t) = \int_0^t \psi(u)\,du \qquad \text{where} \qquad \psi(u) = \begin{cases} u^{p_1-1} & \text{if } 0 \le u \le 1, \\ u^{p_0-1} & \text{if } 1 \le u < \infty. \end{cases}$$

Show that $L^\Psi$ with its norm is equivalent to the space $L^{p_0} + L^{p_1}$.

**25.** Show that a Banach space $\mathcal{B}$ is a Hilbert space if and only if the parallelogram law holds

$$\|f+g\|^2 + \|f-g\|^2 = 2(\|f\|^2 + \|g\|^2).$$

As a consequence, prove that if $L^p(\mathbb{R}^d)$ with the Lebesgue measure is a Hilbert space, then necessarily $p = 2$.

[Hint: For the first part, in the real case, let $(f,g) = \frac{1}{4}(\|f+g\|^2 + \|f-g\|^2)$.]

**26.** Suppose $1 < p_0, p_1 < \infty$ and $1/p_0 + 1/q_0 = 1$ and $1/p_1 + 1/q_1 = 1$. Show that the Banach spaces $L^{p_0} \cap L^{p_1}$ and $L^{q_0} + L^{q_1}$ are duals of each other up to an equivalence of norms. (See Exercise 9 for the relevant definitions of these spaces. Also, Problem $5^*$ gives a generalization of this result.)

**27.** The purpose of this exercise is to prove that the unit ball in $L^p$ is strictly convex when $1 < p < \infty$, in the following sense. Here $L^p$ is the space of real-valued functions whose $p^{\text{th}}$ power are integrable. Suppose $\|f_0\|_{L^p} = \|f_1\|_{L^p} = 1$, and let

$$f_t = (1-t)f_0 + tf_1$$

be the straight-line segment joining the points $f_0$ and $f_1$. Then $\|f_t\|_{L^p} < 1$ for all $t$ with $0 < t < 1$, unless $f_0 = f_1$.

(a) Let $f \in L^p$ and $g \in L^q$, $1/p + 1/q = 1$, with $\|f\|_{L^p} = 1$ and $\|g\|_{L^q} = 1$. Then

$$\int fg\, d\mu = 1$$

only when $f(x) = \operatorname{sign} g(x)|g(x)|^{q-1}$.

(b) Suppose $\|f_{t'}\|_{L^p} = 1$ for some $0 < t' < 1$. Find $g \in L^q$, $\|g\|_{L^q} = 1$, so that

$$\int f_{t'} g\, d\mu = 1$$

and let $F(t) = \int f_t g\, d\mu$. Observe as a result that $F(t) = 1$ for all $0 \le t \le 1$. Conclude that $f_t = f_0$ for all $0 \le t \le 1$.

(c) Show that the strict convexity fails when $p = 1$ or $p = \infty$. What can be said about these cases?

A stronger assertion is given in Problem $6^*$.

[Hint: To prove (a) show that the case of equality in $A^\theta B^{1-\theta} \le \theta A + (1-\theta)B$, for $A, B > 0$ and $0 < \theta < 1$ holds only when $A = B$.]

**28.** Verify the completeness of $\Lambda^\alpha(\mathbb{R}^d)$ and $L^p_k(\mathbb{R}^d)$.

**29.** Consider further the spaces $\Lambda^\alpha(\mathbb{R}^d)$.

(a) Show that when $\alpha > 1$ the only functions in $\Lambda^\alpha(\mathbb{R}^d)$ are the constants.

(b) Motivated by (a), one defines $C^{k,\alpha}(\mathbb{R}^d)$ to be the class of functions $f$ on $\mathbb{R}^d$ whose partial derivatives of order less than or equal to $k$ belong to $\Lambda^\alpha(\mathbb{R}^d)$. Here $k$ is an integer and $0 < \alpha \leq 1$. Show that this space, endowed with the norm

$$\|f\|_{C^{k,\alpha}} = \sum_{|\beta| \leq k} \left\| \partial_x^\beta f \right\|_{\Lambda^\alpha(\mathbb{R}^d)},$$

is a Banach space.

**30.** Suppose $\mathcal{B}$ is a Banach space and $\mathcal{S}$ is a closed linear subspace of $\mathcal{B}$. The subspace $\mathcal{S}$ defines an equivalence relation $f \sim g$ to mean $f - g \in \mathcal{S}$. If $\mathcal{B}/\mathcal{S}$ denotes the collection of these equivalence classes, then show that $\mathcal{B}/\mathcal{S}$ is a Banach space with norm $\|f\|_{\mathcal{B}/\mathcal{S}} = \inf(\|f'\|_{\mathcal{B}}, \quad f' \sim f)$.

**31.** If $\Omega$ is an open subset of $\mathbb{R}^d$ then one definition of $L^p_k(\Omega)$ can be taken to be the quotient Banach space $\mathcal{B}/\mathcal{S}$, as defined in the previous exercise, with $\mathcal{B} = L^p_k(\mathbb{R}^d)$ and $\mathcal{S}$ the subspace of those functions which vanish a.e. on $\Omega$. Another possible space, that we will denote by $L^p_k(\Omega^0)$, consists of the closure in $L^p_k(\mathbb{R}^d)$ of all $f$ that have compact support in $\Omega$. Observe that the natural mapping of $L^p_k(\Omega^0)$ to

$L^p_k(\Omega)$ has norm equal to 1. However, this mapping is in general not surjective. Prove this in the case when $\Omega$ is the unit ball and $k \geq 1$.

**32.** A Banach space is said to be separable if it contains a countable dense subset. In Exercise 11 we saw an example of a Banach space $\mathcal{B}$ that is separable, but where $\mathcal{B}^*$ is not separable. Prove, however, that in general when $\mathcal{B}^*$ is separable, then $\mathcal{B}$ is separable. Note that this gives another proof that in general $L^1$ is not the dual of $L^\infty$.

**33.** Let $V$ be a vector space over the complex numbers $\mathbb{C}$, and suppose there exists a real-valued function $p$ on $V$ satisfying:

$$\begin{cases} p(\alpha v) = |\alpha| p(v), & \text{if } \alpha \in \mathbb{C}, \text{ and } v \in V, \\ p(v_1 + v_2) \leq p(v_1) + p(v_2), & \text{if } v_1 \text{ and } v_2 \in V. \end{cases}$$

Prove that if $V_0$ is a subspace of $V$ and $\ell_0$ a linear functional on $V_0$ which satisfies $|\ell_0(f)| \leq p(f)$ for all $f \in V_0$, then $\ell_0$ can be extended to a linear functional $\ell$ on $V$ that satisfies $|\ell(f)| \leq p(f)$ for all $f \in V$.

[Hint: If $u = \mathrm{Re}(\ell_0)$, then $\ell_0(v) = u(v) - iu(iv)$. Apply Theorem 5.2 to $u$.]

**34.** Suppose $\mathcal{B}$ is a Banach space and $\mathcal{S}$ a closed proper subspace, and assume $f_0 \notin \mathcal{S}$. Show that there is a continuous linear functional $\ell$ on $\mathcal{B}$, so that $\ell(f) = 0$ for $f \in \mathcal{S}$, and $\ell(f_0) = 1$. The linear functional $\ell$ can be chosen so that $\|\ell\| = 1/d$ where $d$ is the distance from $f_0$ to $\mathcal{S}$.

**35.** A linear functional $\ell$ on a Banach space $\mathcal{B}$ is continuous if and only if $\{f \in \mathcal{B} : \ell(f) = 0\}$ is closed.

[Hint: This is a consequence of Exercise 34.]

**36.** The results in Section 5.4 can be extended to $d$-dimensions.

(a) Show that there exists an extended-valued non-negative function $\hat{m}$ defined on all subsets of $\mathbb{R}^d$ so that (i) $\hat{m}$ is finitely additive; (ii) $\hat{m}(E) = m(E)$ whenever $E$ is Lebesgue measurable, where $m$ is Lebesgue measure; and $\hat{m}(E+h) = \hat{m}(E)$ for all sets $E$ and every $h \in \mathbb{R}^d$. Prove this is as a consequence of (b) below.

(b) Show that there is an "integral" $I$, defined on all bounded functions on $\mathbb{R}^d/\mathbb{Z}^d$, so that $I(f) \geq 0$ whenever $f \geq 0$; the map $f \mapsto I(f)$ is linear; $I(f) = \int_{\mathbb{R}^d/\mathbb{Z}^d} f\,dx$ whenever $f$ is measurable; and $I(f_h) = I(f)$ where $f_h(x) = f(x-h)$, and $h \in \mathbb{R}^d$.

## Chapter 2. $L^p$ Spaces in Harmonic Analysis

**1.** Show that an inequality

$$\|\{a_n\}\|_{L^q} \le A\|f\|_{L^p}, \qquad \text{for all } f \in L^p,$$

with $a_n = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-in\theta}\,d\theta$, is possible only if $1/p + 1/q \le 1$.
[Hint: Let $D_N(\theta) = \sum_{|n|\le N} e^{in\theta}$ be the Dirichlet kernel. Then $\|D_N\|_{L^p} \approx N^{1-1/p}$ as $N \to \infty$, if $p > 1$ and $\|D_N\|_{L^1} \approx \log N$.]

**2.** The following are simple generalizations of the Hausdorff-Young inequalities.

(a) Suppose $\{\varphi_n\}$ is an orthonormal sequence on $L^2(X,\mu)$. Assume also that $|\varphi_n(x)| \le M$ for all $n$. If $a_n = \int f\overline{\varphi_n}\,d\mu$, then $\|a_n\|_{L^q} \le M^{(2/p)-1}\|f\|_{L^p(X)}$, $1 \le p \le 2$, $1/p + 1/q = 1$.

(b) Suppose $f \in L^p$ on the torus $\mathbb{T}^d$, and $a_n = \int_{\mathbb{T}^d} f(x)e^{-2\pi i n\cdot x}\,dx$, $n \in \mathbb{Z}^d$. Then $\|\{a_n\}\|_{L^q(\mathbb{Z}^d)} \le \|f\|_{L^p(\mathbb{T}^d)}$, where $1/q \le 1 - 1/p$.

**3.** Check that an inequality of the form $\|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A\|f\|_{L^p(\mathbb{R}^d)}$ (holding for all simple functions $f$) is possible if and only if $1/p + 1/q = 1$.
[Hint: Let $f_r(x) = f(rx)$, $r > 0$. Then $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$.]

**4.** Prove that another necessary condition for the inequality in the previous exercise is that $p \leq 2$. In fact the estimate

$$\int_{|\xi|\leq 1} |\hat{f}(\xi)|\, d\xi \leq A\|f\|_{L^p}$$

can hold only if $p \leq 2$.
[Hint: Let $f^s(x) = s^{-d/2}e^{-\pi|x|^2/s}$, $s = \sigma + it$, $\sigma > 0$. Then $(\widehat{f^s})(\xi) = e^{-\pi s|\xi|^2}$. Note that $\|f^s\|_{L^p} \leq ct^{d(1/p-1/2)}$ when $\sigma = 1$, and let $t \to \infty$.]

**5.** Let $\psi$ be the conformal map of the strip $0 < \mathrm{Re}(z) < 1$ to the upper half-plane defined by $\psi(z) = e^{i\pi z}$. Check that $\Phi(z) = e^{-i\psi(z)}$ is continuous on the closure of the strip, $|\Phi(z)| = 1$ on the boundary lines, but $\Phi(z)$ is unbounded in the strip.

**6.** Extend the Riesz convexity theorem (in Section 2) to the $L^{p,r}$ spaces discussed in Exercise 18 of Chapter 1. We assume $T$ is a linear transformation from simple functions to locally integrable functions. Suppose

$$\|T(f)\|_{L^{q_0,s_0}} \leq M_0\|f\|_{L^{p_0,r_0}}, \qquad \text{and} \qquad \|T(f)\|_{L^{q_1,s_1}} \leq M_1\|f\|_{L^{p_1,r_1}}$$

for all simple $f$. Prove that as a consequence $\|T(f)\|_{L^{q,s}} \leq M_\theta\|f\|_{L^{p,r}}$ where $\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$, $\frac{1}{r} = \frac{1-\theta}{r_0} + \frac{\theta}{r_1}$, $\frac{1}{q} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}$, $\frac{1}{s} = \frac{1-\theta}{s_0} + \frac{\theta}{s_1}$, and $0 \leq \theta \leq 1$.

[Hint: Suppose $f$ and $g$ are a pair of simple functions with $\|f\|_{L^{p,r}} \leq 1$ and $\|f\|_{L^{q',s'}} \leq 1$. Define

$$f_z = |f(x,t)|^{p\alpha(z)} \frac{f(x,t)}{|f(x,t)|} \|f(\cdot,t)\|_{L^p(dx)}^{r\beta(z)-p\alpha(z)},$$

where $\alpha(z) = \frac{1-z}{p_0} + \frac{z}{p_1}$, $\beta(z) = \frac{1-z}{r_0} + \frac{z}{r_1}$. Note that when $z = \theta$, then $f_z = f$. Also

$$\|f_{1+it}\|_{L^{p_1,r_1}} \leq 1 \qquad \text{and} \qquad \|f_{0+it}\|_{L^{p_0,r_0}} \leq 1.$$

Make an analogous definition for $g_z$ and consider $\int T(f_z)g_z\,dxdt$.]

**7.** Suppose $f$ is a bounded function on $\mathbb{R}$ with compact support. Then $H(f) \in L^1(\mathbb{R})$ if and only if $\int f\,dx = 0$.
[Hint: If $a = \int f\,dx$, then $H(f)(x) = \frac{a}{\pi x} + O(1/x^2)$ as $|x| \to \infty$.]

**8.** Suppose $T$ is a bounded linear transformation mapping the space of real-valued $L^p$ functions into itself with

$$\|T(f)\|_{L^p} \leq M\|f\|_{L^p}.$$

(a) Let $T'$ be the extension of $T$ to complex-valued functions: $T'(f_1 + if_2) = T(f_1) + iT(f_2)$. Then $T'$ has the same bound: $\|T'(f)\|_{L^p} \leq M\|f\|_{L^p}$.

(b) More generally, fix any $N$, then

$$\|(\sum_{j=1}^{N} |T(f_j)|^2)^{1/2}\|_{L^p} \leq M\|(\sum_{j=1}^{N} |f_j|^2)^{1/2}\|_{L^p}.$$

[Hint: For part (b), let $\xi$ denote a unit vector in $\mathbb{R}^N$, and let $F_\xi = \sum_{j=1}^{N} \xi_j f_j$, $\xi = (\xi_1, \ldots, \xi_N)$. Then $\int |(TF_\xi)(x)|^p \leq M^p \int |F_\xi(x)|^p$. Integrate this inequality for $\xi$ on the unit sphere.]

**9.** Show the identity of the following two classes of harmonic functions $u$ in the upper half-plane $\mathbb{R}^2_+ = \{z : \ x + iy, \ y > 0\}$.

(a) The harmonic functions $u$ that are continuous in the closure $\overline{\mathbb{R}^2_+}$, and that vanish at infinity (that is, $u(x, y) \to 0$, if $|x| + y \to \infty$).

(b) The functions representable as $u(x, y) = (f * \mathcal{P}_y)(x)$, where $\mathcal{P}_y(x)$ is the Poisson kernel $\frac{1}{\pi} \frac{y}{x^2+y^2}$, and $f$ is a continuous function on $\mathbb{R}$ that vanishes at infinity (that is, $f(x) \to 0$, as $|x| \to \infty$).

[Hint: To show that (a) implies (b), let $f(x) = u(x, 0)$. Then $\mathcal{D}(x, y) = u(x, y) - (f * \mathcal{P}_y)(x)$ is harmonic in $\overline{\mathbb{R}^2_+}$, continuous on $\overline{\mathbb{R}^2_+}$, vanishes at infinity, and moreover $\mathcal{D}(x, 0) = 0$. Thus by the maximum principle, $\mathcal{D}(x, y) = 0$.]

**10.** Suppose $f \in L^p(\mathbb{R})$. Verify that:

(a) $\|f * \mathcal{P}_y\|_{L^p(\mathbb{R})} \leq \|f\|_{L^p}$, $1 \leq p \leq \infty$.

(b) $f * \mathcal{P}_y \to f$, as $y \to 0$, in the $L^p$ norm, when $1 \leq p < \infty$.

**11.** Assume $f \in L^p(\mathbb{R})$, $1 < p < \infty$. Prove that:

(a) $f * \mathcal{Q}_y = H(f) * \mathcal{P}_y$, where $H$, $\mathcal{P}_y$ and $\mathcal{Q}_y$ are respectively the Hilbert transform, the Poisson kernel and the conjugate Poisson kernel.

(b) $f * \mathcal{Q}_y \to H(f)$ in the $L^p$ norm, as $y \to 0$.

(c) $H_\epsilon(f) \to H(f)$ in the $L^p$ norm, as $\epsilon \to 0$.

[Hint: Verify (a) first for $f \in L^2$ by noting that the Fourier transform of both sides equals $\hat{f}(\xi)\frac{\text{sign}(\xi)}{\xi}e^{-2\pi|\xi|y}$.]

**12.** In $\mathbb{R}^d$, suppose $f(x) = |x|^{-d}(\log 1/|x|)^{-1-\delta}$ if $|x| \leq 1/2$, $f(x) = 0$ otherwise. Then observe that $f^*(x) \geq c|x|^{-d}(\log 1/|x|)^{-\delta}$, if $|x| \leq 1/2$. Hence if $0 < \delta \leq 1$, we have $f \in L^1(\mathbb{R}^d)$ but $f^*(x)$ is not integrable over the unit ball.

**13.** Prove that the basic distribution function inequality (28) for the maximal function can essentially be reversed, that is, there is a constant $A$ so that

$$m(\{x:\ f^*(x) > \alpha\}) \geq (A/\alpha) \int_{|f(x)|>\alpha} |f(x)|\, dx.$$

[Hint: Write $E_\alpha = \{x:\ f^*(x) > \alpha\}$ as $\bigcup_{j=1}^{\infty} Q_j$, with $Q_j$ closed cubes satisfying (37), with $\Omega = E_\alpha$. For each $Q_j$ let $B_j$ be the smallest ball so that $Q_j \subset B_j$, and $\overline{B_j}$ intersects $E_\alpha^c$. First $m(B_j) \leq cm(Q_j)$, then $\frac{1}{m(B_j)} \int_{B_j} |f|\, dx \leq \alpha$. Thus $m(Q_j) \geq \frac{c^{-1}}{\alpha} \int_{B_j} |f(x)|\, dx \geq \frac{c^{-1}}{\alpha} \int_{Q_j} |f(x)|\, dx$. Now add in $j$, and use the fact that $\{x:\ |f(x)| > \alpha\} \subset \{x:\ f^*(x) > \alpha\}$.]

**14.** Deduce the following important consequence from (28) and the previous exercise. Suppose $f$ is an integrable function on $\mathbb{R}^d$, and $B_1$, $B_2$ are a pair of balls with $\overline{B_1} \subset B_2$.

(a) $f^*$ is integrable on $B_1$ if $|f| \log(1 + |f|)$ is integrable on $B_2$.

(b) In the converse direction, whenever $f^*$ is integrable on $B_1$ then $|f| \log(1 + |f|)$ is also integrable there.

[Hint: Integrate the inequalities in $\alpha$, for $\alpha \geq 1$.]

**15.** Consider the weak-type space, consisting of all functions $f$ for which $m(\{x : |f(x)| > \alpha\}) \leq \frac{A}{\alpha}$ for some $A$ and all $\alpha > 0$. One might hope to define a norm on this space by taking the "norm" of $f$ to be the least $A$ for which the above inequality holds. Denote this quantity by $\mathcal{N}(f)$.

(a) Show, however, that $\mathcal{N}$ is not a genuine norm; moreover there is no norm $\|\cdot\|$ on this space so that $\|f\|$ is equivalent with $\mathcal{N}(f)$.

(b) Prove also that this space has no non-trivial bounded linear functionals.

[Hint: Consider $\mathbb{R}$. The function $f(x) = 1/|x|$ has $\mathcal{N}(f) = 2$. But if $f_N(x) = \frac{1}{N}[f(x+1) + f(x+2) + \cdots + f(x+N)]$, then $\mathcal{N}(f_N) \geq c \log N$.]

**16.** Prove that the space $\mathbf{H}^1_r$ is complete as follows. Let $\{f_n\}$ be a Cauchy sequence in $\mathbf{H}^1_r$. Then since $\{f_n\}$ is also Cauchy in $L^1$, there is an $L^1$ function $f$ so that $f = \lim_{n\to\infty} f_n$ in the $L^1$ norm. Now for an appropriate sub-sequence $\{n_k\}$, write $f = f_{n_1} + \sum_{k=1}^{\infty}(f_{n_{k+1}} - f_{n_k})$.

**17.** Consider the function $f$ defined by $f(x) = 1/(x(\log x)^2)$ for $0 < x \leq 1/2$ and $f(x) = 0$ if $x > 1/2$, and extended to $x < 0$ by $f(x) = -f(-x)$. Then $f$ is integrable on $\mathbb{R}$, with $\int f = 0$, hence $f$ is a multiple of a 1-atom in the terminology of Section 5.2.

Verify that $M(f) \geq c/(|x| \log |x|)$ for $|x| \leq 1/2$, hence $M(f) \notin L^1$, thus by Theorem 6.1 we know that $f \notin \mathbf{H}^1_r$.

**18.** Show that there exists a $c > 1$ so that every $f \in L^1(\mathbb{R}^d)$ can be written as $f(x) = \sum_{k=1}^{\infty} \lambda_k \mathfrak{a}_k(x)$, with $\sum |\lambda_k| \leq c\|f\|_{L^1}$, where the $\mathfrak{a}_k$ are "faux" atoms: each $\mathfrak{a}_k$ is supported in a ball $B_k$; $|\mathfrak{a}_k(x)| \leq 1/m(B_k)$ for all $x$; but $\mathfrak{a}_k$ does not necessarily satisfy the cancelation condition $\int \mathfrak{a}_k(x)\,dx = 0$.

[Hint: Let $f_n = \mathbb{E}_n(f)$, where $\mathbb{E}_n$ replaces $f$ by its average over each dyadic cube of the $n^{\text{th}}$-generation. Then $\|f_n - f\|_{L^1} \to 0$. Pick $\{n_k\}$ so that $\|f_{n_{k+1}} - f_{n_k}\|_{L^1} < 1/2^k$, and write $f = f_{n_1} + \sum_{k=1}^{\infty}(f_{n_{k+1}} - f_{n_k})$.]

**19.** The following illustrates two senses in which $\mathbf{H}^1_r$ is near $L^1$, but yet different.

(a) Suppose $f_0(x)$ is a positive decreasing function on $(0,\infty)$ that is integrable on $(0,\infty)$. Then show that there is a function $f \in \mathbf{H}^1_r(\mathbb{R})$ so that $|f(x)| \geq f_0(|x|)$.

(b) However if $f \in \mathbf{H}^1_r(\mathbb{R}^d)$, and $f$ is positive on an open set, then its size must be "smaller" on that open set than a general integrable function. In fact, prove that if $f \in \mathbf{H}^1_r$, and $f \geq 0$ in a ball $B_1$, then $f \log(1+f)$ must be integrable over any proper sub-ball $B_0 \subset B_1$.

[Hint: For (a) take $f(x) = \text{sign}(x) f_0(|x|)$, and find an atomic decomposition for $f$. For (b) use Exercise 14, together with the maximal theorem in Section 6, with $\Phi$ positive.]

**20.** When $f \in L^1(\mathbb{R}^d)$ we know that its Fourier transform $\hat{f}$ is bounded and $\hat{f}(\xi)$ tends to 0 as $|\xi| \to \infty$ (the Riemann-Lebesgue lemma), but no better assertion about the "smallness" of $\hat{f}$ can be made. (For the analogous result for Fourier series, see Chapter 3 in Book III.) Show, however, that for $f \in \mathbf{H}^1_r$ we have

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \leq A\|f\|_{\mathbf{H}^1_r}.$$

[Hint: Verify this for atoms.]

**21.** Prove that if $|f(x)| \leq A(1+|x|)^{-d-1}$, and $\int_{\mathbb{R}^d} f(x)\,dx = 0$, then $f \in \mathbf{H}^1_r(\mathbb{R}^d)$. [Hint: While this is elementary, it is a little tricky. Write $f = \sum_{k=0}^{\infty} f_k$, where $f_0(x) = f(x)$ if $|x| \leq 1$, 0 elsewhere, and $f_k(x) = f(x)$ if $2^{k-1} < |x| \leq 2^k$ and 0 elsewhere, and $k \geq 1$. Let $c_k = \int f_k\,dx$, $s_k = \sum_{j \geq k} c_j$, then $s_0 = 0$. Fix a bounded function $\eta$ supported in $|x| \leq 1$, with $\int \eta(x)\,dx = 1$. Now write $f(x) = \sum_{k=0}^{\infty}(f_k - c_k\eta_k) + \sum_{k=0}^{\infty} c_k\eta_k$, where $\eta_k(x) = 2^{-kd}\eta(2^{-k}x)$ and $\int \eta_k = 1$. The first sum is clearly a sum of multiples of atoms (which are $O(2^{-k})$) supported on the balls $|x| \leq 2^k$. That the second sum is similar can be seen by rewriting it as $\sum_{k=1}^{\infty} s_k(\eta_k - \eta_{k-1})$.]

**22.** Let $f$ be the atom on $\mathbb{R}$ supported in $|x| \leq 1/2$ given by $f(x) = \text{sign}(x)$. Apply to $f$ the maximal function $f_0^*$ defined by

$$f_0^*(x) = \sup_{\epsilon>0} |(f * \chi_\epsilon)(x)|,$$

where $\chi$ is the characteristic function of $|x| \leq 1/2$ and $\chi_\epsilon(x) = \epsilon^{-1}\chi(x/\epsilon)$.

Verify that $|f_0^*(x)| \geq 1/(2|x|)$ if $|x| \geq 1/2$ hence $f_0^* \notin L^1$. Thus the maximal function $f_0^*$, defined in terms of $\chi$, cannot be used to characterize the real Hardy space $\mathbf{H}_r^1$.

**23.** Verify the following examples related to BMO:

(a) $\log|x| \in \mathrm{BMO}(\mathbb{R}^d)$.

(b) If $f(x) = \log x$, when $x > 0$, and $= 0$ when $x \leq 0$, then $f \notin \mathrm{BMO}(\mathbb{R})$.

(c) If $\delta \geq 0$, $(\log|x|)^\delta \in \mathrm{BMO}(\mathbb{R}^d)$ if and only if $\delta \leq 1$.

[Hint: With $f(x) = \log|x|$, note that $f(rx) = f(x) + c_r$ and so we may assume the ball $B$ has radius 1 in testing the condition (43). For (b), test $f$ on small intervals centered at the origin.]

**24.** Using Exercises 19 (a) and 23, give examples of $f \in \mathbf{H}^1_r$ and $g \in \mathrm{BMO}$ so that $|f(x)g(x)|$ is not integrable over $\mathbb{R}^d$.

## Chapter 3. Distributions: Generalized Functions

**1.** Suppose $F$ is a distribution on $\Omega$ and $F = f$, with $f$ a $C^k$ function in $\Omega$. Show that $\partial_x^\alpha F$, taken in the sense of distributions, agrees with $\partial_x^\alpha f$ for each $|\alpha| \leq k$.

**2.** The following represent converses to the previous exercise.

(a) Suppose $f$ and $g$ are continuous functions on $(a,b) \subset \mathbb{R}$ and $\frac{df}{dx}$ (taken in the sense of distributions) agrees with $g$. Show that for every $x \in (a,b)$, $(f(x+h) - f(x))/h \to g(x)$ as $h \to 0$.

(b) If $f$ and $g$ are merely assumed to be in $L^1(a,b)$ with $\frac{df}{dx} = g$ in the sense of distributions, then $f$ is absolutely continuous and $(f(x+h) - f(x))/h \to g(x)$ as $h \to 0$ for a.e. $x$.

As a result, if $f$ is a continuous but nowhere differentiable function on $\mathbb{R}$, then the distribution derivative of $f$ is not a locally integrable function on any sub-interval.

(c) Generalize (a) as follows: Suppose $k \geq 1$ is an integer, and that $f$ is a continuous function on an open set $\Omega$. If for each multi-index $\alpha$ with $|\alpha| \leq k$, the distribution $\partial_x^\alpha f$ equals a continuous function $g_\alpha$, then $f$ is of class $C^k$ and $\partial_x^\alpha f = g_\alpha$ as functions, for all $|\alpha| \leq k$.

[Hint: To see (a), let $x_0 \in (a,b)$, $h > 0$, and let $\eta$ be a test function on $(a,b)$ so that $\int \eta = 1$. With $\delta > 0$, define $\eta^\delta(x) = \delta^{-1}\eta(x/\delta)$ and

$$\varphi(x) = \int_{-\infty}^{x} \eta^\delta(x_0 + h - y) - \eta^\delta(x_0 - y)\, dy.$$

Then $\int f(x) \frac{d}{dx}\varphi(x)\, dx = -\int g(x)\varphi(x)\, dx$ and let $\delta, h \to 0$.

For (b), show as a first step that, up to a constant, $f$ equals the indefinite integral of $g$, almost everywhere. Then use Theorem 3.8 in Chapter 3, Book III, about the differentiability almost everywhere of an absolutely continuous function.]

**3.** Show that a bounded function $f$ on $\mathbb{R}^d$ satisfies a Lipschitz condition (also known as a Hölder condition of exponent 1)

$$|f(x) - f(y)| \leq C|x - y|, \qquad \text{for all } x, y \in \mathbb{R}^d,$$

if and only if $f \in L^\infty$ and all the first order partial derivatives $\partial f/\partial x^j$, $1 \leq j \leq d$, belong to $L^\infty$ in the sense of distributions.

[Hint: Let $f_n = f * \psi_n$, where $\psi_n$ is an approximation to the identity as in Corollary 1.2. Then $\partial f_n/\partial x_j \in L^\infty$ uniformly in $n$.]

**4.** Suppose $F$ is a distribution on $\Omega$.

(a) There exist $f_n \in C^\infty$, each of compact support in $\Omega$, so that $f_n \to F$ in the sense of distributions.

(b) If $F$ is supported in the compact set $C$, then for every $\epsilon > 0$ we can choose the $f_n$ so that their supports are in the $\epsilon$-neighborhood of $C$.

**5.** Let $f$ be locally integrable on $\mathbb{R}^d$. Then the "support" of $f$ in the measure-theoretic sense is the set $E = \{x : f(x) \neq 0\}$. Note that $E$ is essentially determined only modulo sets of measure zero.

Show that the support of $f$, as a distribution, is equal to the intersection of all closed sets $C$ such that $E - C$ has measure zero.

**6.** Assume that $\Omega$ is a region in $\mathbb{R}^d$ defined by $\Omega = \{x \in \mathbb{R}^d : \ x_d > \varphi(x')\}$, with $x = (x', x_d) \in \mathbb{R}^{d-1} \times \mathbb{R}$, and $\varphi$ a $C^1$ function. Suppose $f$ is a function that is continuous in $\overline{\Omega}$ and whose first derivatives are also continuous in $\overline{\Omega}$, with $f|_{\partial\Omega} = 0$.

Let $\tilde{f}$ be the extension of $f$ to $\mathbb{R}^d$ defined by $\tilde{f}(x) = f(x)$ if $x \in \overline{\Omega}$, and $\tilde{f}(x) = 0$ if $x \notin \overline{\Omega}$. Then $\frac{\partial \tilde{f}}{\partial x_j}$, taken in the sense of distributions, is the function which is $\frac{\partial f}{\partial x_j}$ in $\overline{\Omega}$, and zero in $\overline{\Omega}^c$. (Note that it is not necessarily true that $\frac{\partial \tilde{f}}{\partial x_j}$ is continuous.) [Hint: Show that $-\int_\Omega f(x) \frac{\partial \psi}{\partial x_j}\, dx = \int_\Omega \frac{\partial f}{\partial x_j} \psi \, dx$ for all $C^\infty$ functions $\psi$ of compact support in $\mathbb{R}^d$.]

**7.** Show that the distribution $F$ is tempered if and only if there is an integer $N$, and a constant $A$, so that for all $R \geq 1$,

$$|F(\varphi)| \leq AR^N \sup_{|x|\leq R,\ 0\leq|\alpha|\leq N} |\partial_x^\alpha \varphi(x)|\,,$$

for all $\varphi \in \mathcal{D}$ supported in $|x| \leq R$.

**8.** Suppose $F$ is a homogeneous distribution of degree $\lambda$. Show that $F$ is tempered.

[Hint: Fix $\eta \in \mathcal{D}$, $\eta(x) = 1$, for $|x| \leq 1$, $\eta$ supported in $|x| \leq 2$. Let $\eta_R(x) = \eta(x/R)$. Find $N$ so that $|\eta_1 F(\varphi)| \leq c\|\varphi\|_N$. Then deduce that $|(\eta_R F)(\varphi)| \leq cR^{N+|\lambda|}\|\varphi\|_N$.]

**9.** Check that on the real line, $f(x) = e^x$, considered as a distribution, is not tempered.

[Hint: Show that the criterion in Exercise 7 fails for every $N$.]

**10.** Verify that $\mathcal{D}$ is dense in $\mathcal{S}$.
[Hint: Fix $\eta \in \mathcal{D}$ so that $\eta = 1$ in a neighborhood of the origin. Let $\eta_k(x) = \eta(x/k)$ and consider $\varphi_k = \eta_k \varphi$.]

**11.** Suppose that $\varphi_1, \varphi_2 \in \mathcal{S}$.

(a) Verify that $\varphi_1 \cdot \varphi_2$ belongs to $\mathcal{S}$.

(b) Using the Fourier transform, prove that $\varphi_1 * \varphi_2 \in \mathcal{S}$.

(c) Show directly from the definition of convolution that $\varphi_1 * \varphi_2 \in \mathcal{S}$.

**12.** Prove that if $F_1$ is a distribution of compact support and $\varphi \in \mathcal{S}$, then $F_1 * \varphi \in \mathcal{S}$.

[Hint: For each $N$, there exists a constant $c_N$ so that

$$\|\psi_y^{\sim}\|_N \leq c_N(1+|y|)^N \|\psi\|_N.]$$

**13.** Use the previous exercise to prove that if $F_1$ and $F$ are distributions with $F_1$ having compact support and $F$ being tempered then:

(a) $F * F_1$ is tempered, and;

(b) $(F * F_1)^\wedge = F_1^\wedge F^\wedge$, ($F_1^\wedge$ is $C^\infty$ and slowly increasing.)

**14.** Check that $f(x) = \frac{1}{2}|x|$ is a fundamental solution for $\frac{d^2}{dx^2}$ on $\mathbb{R}$.

**15.** A $d$-dimensional generalization of the identity for the Heaviside function is the identity

$$\delta = \sum_{j=1}^{d} \left( \frac{\partial}{\partial x_j} \right) h_j,$$

with $h_j(x) = \frac{1}{A_d} \frac{x_j}{|x|^d}$, and $A_d = 2\pi^{d/2}/\Gamma(d/2)$ denotes the area of the unit sphere in $\mathbb{R}^d$.
[Hint: When $d > 2$, write $\delta = \sum_{j=1}^{d} \frac{\partial}{\partial x_j} \left( \frac{\partial}{\partial x_j} C_d |x|^{-d+2} \right)$.]

**16.** Consider the complex plane $\mathbb{C} = \mathbb{R}^2$, with $z = x + iy$.

(a) Note that the Cauchy-Riemann operator

$$\partial_{\overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

is elliptic.

(b) Show that the locally integrable function $1/(\pi z)$ is a fundamental solution for $\partial_{\overline{z}}$.

(c) Suppose $f$ is continuous in $\Omega$, and $\partial_{\overline{z}} f = 0$ in the sense of distributions. Then $f$ is analytic.

[Hint: For (b), use Theorem 2.9, and note that $\triangle = 4\partial_{\overline{z}}\partial_z$, where $\partial_z = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$.]

**17.** Suppose $f(z)$ is a meromorphic function on $\Omega \subset \mathbb{C}$. Prove:

(a) $\log|f(z)|$ is locally integrable.

(b) $\triangle(\log|f(z)|)$ taken in the sense of distributions is equal to $2\pi\sum_j m_j\delta_j - 2\pi\sum_k m'_k\delta_k$. Here the $\delta_j$ are the delta functions placed at the distinct zeroes of $f$, namely $\delta_j(\varphi) = \varphi(z_j)$, and the $\delta_k$ are placed at the poles $z'_k$ of $f$; also $m_j$, and $m'_k$ are the respective multiplicities.

[Hint: $\frac{1}{2\pi}\log|z|$ is a fundamental solution of $\triangle$.]

**18.** Prove that a distribution $F$ is homogeneous of degree $\lambda$ if and only if

$$\sum_{j=1}^{d} x_j \frac{\partial F}{\partial x_j} = \lambda F.$$

[Hint: For the converse, consider $\Phi(a) = F(\varphi^a)$ for $a > 0$, $\varphi \in \mathcal{D}$. Then $\Phi(a)$ is $C^\infty$ for $a > 0$, and $\frac{d\Phi(a)}{da} = \frac{\lambda}{a}\Phi(a)$.]

**19.** Prove the following facts about distributions in $\mathbb{R}$.

(a) Given a distribution $F$, there exists a distribution $F_1$ so that

$$\frac{d}{dx}F_1 = F.$$

(b) Show that $F_1$ is unique modulo an additive constant.

[Hint: For (a) fix $\varphi_0 \in \mathcal{D}$, with $\int \varphi_0 = 1$, and note that each $\varphi \in \mathcal{D}$ can be written uniquely as $\varphi = \frac{d\psi}{dx} + a\varphi_0$ for some $\psi \in \mathcal{D}$ and a constant $a$. Then define $F_1(\varphi) = F(\psi)$. For (b), use the fact that $d/dx$ is elliptic.]

**20.** Show that if $\lambda_1, \ldots, \lambda_d$ are distinct complex exponents and $\sum_{j=1}^{n}(a_j x^{\lambda_j} + b_j x^{\lambda_j} \log x) = 0$ for all $x > 0$, then $a_j = b_j = 0$ for all $1 \leq j \leq n$.

[Hint: Proceed as in the proof of Lemma 2.6, and use the fact that $\int_1^R x^{-1+i\mu_j}\, dx$ is equal to $\log R$ if $\mu_j = 0$ and that this integral is $O(1)$ if $\mu_j$ is real and $\neq 0$.]

**21.** Let $F(x,t) = \mathcal{H}_t(x)$, for $t > 0$, and $F(x,t) = 0$, when $t \leq 0$, as in Theorem 2.10. Prove directly that

$$\hat{F}(\xi,\tau) = \frac{1}{4\pi^2|\xi|^2 + 2\pi i\tau}$$

where $(\xi,\tau) \in \mathbb{R}^d \times \mathbb{R}$, with $\xi$ dual to $x$, and $\tau$ dual to $t$.
[Hint: Use the two identities

$$\int_0^\infty e^{-4\pi^2|\xi|^2 t} e^{-2\pi i\tau t}\, dt = \frac{1}{4\pi^2|\xi|^2 + 2\pi i\tau} \qquad \text{for } |\xi| > 0$$

and

$$\int_{\mathbb{R}^d} \mathcal{H}_t(x) e^{-2\pi i x\cdot\xi}\, dx = e^{-4\pi^2|\xi|^2 t} \qquad \text{for } t > 0.]$$

**22.** Suppose $f$ is a locally integrable function defined on $\mathbb{R}$, and let $u$ be the function defined by $u(x,t) = f(x-t)$, for $(x,t) \in \mathbb{R}^2$. Verify that $u$, taken as a distribution, satisfies the wave equation

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}.$$

More generally, let $F$ be any distribution on $\mathbb{R}$. Construct $U$ (in analogy to $f(x-t)$) as follows. If $\varphi$ is in $\mathcal{D}(\mathbb{R}^2)$, $\mathbb{R}^2 = \{(x,t)\}$, set $U(\varphi) = \int_{\mathbb{R}} (F * \varphi(x,\cdot))(x)\,dx$. Then $U$ satisfies

$$\frac{\partial^2 U}{\partial x^2} = \frac{\partial^2 U}{\partial t^2}.$$

Note that $U$ is invariant under the translations $(h,h)$, for $h \in \mathbb{R}$.

**23.** Show that in $\mathbb{R}^3$ the function

$$F(x) = \frac{-1}{4\pi|x|} e^{-|x|}$$

is a fundamental solution of the operator $\triangle - I$. The function $F$ is the "Yukawa potential" in the theory of elementary particles. In contrast to the "Newtonian potential" $-1/(4\pi|x|)$, the fundamental solution of $\triangle$, the function $F$ has a very rapid decay at infinity and it thus accounts for the short-range forces in the theory. [Hint: Let $F$ be the inverse Fourier transform of $-(1 + 4\pi^2|\xi|^2)^{-1}$. Going to polar coordinates in $\mathbb{R}^3$, one then uses the identity

$$\int_{|\xi|=1} e^{2\pi i \xi \cdot x} \, d\sigma(\xi) = \frac{2\sin(2\pi|x|)}{|x|},$$

together with the Fourier transform of the conjugate Poisson kernel, given by (18) of the previous chapter.]

**24.** The following statements deal with the uniqueness of the fundamental solutions of the Laplacian.

(a) Up to an additive constant, the unique fundamental solutions of $\triangle$ in $\mathbb{R}^d$, $d \geq 2$, that are rotationally invariant, are the ones given in Theorems 2.8 and 2.9.

(b) The unique fundamental solution of $\triangle$ in $\mathbb{R}^d$, $d \geq 3$, that vanishes at infinity is the one given in Theorem 2.8.

**25.** A distribution $F$ defined on $\Omega \subset \mathbb{R}$ is **positive** if $F(\varphi) \geq 0$ for all $\varphi \in \mathcal{D}$ supported in $\Omega$, with $\varphi \geq 0$. Show that $F$ is positive if and only if $F(\varphi) = \int \varphi \, d\mu$ for some Borel measure $d\mu$ on $\Omega$ that is finite on compact subsets.

**26.** Recall that a real-valued function on $(a, b)$ is **convex** if $f(x_0(1-t)+x_1t) \leq (1-t)f(x_0)+tf(x_1)$, for $x_0, x_1 \in (a, b)$, $0 \leq t \leq 1$. (See also Problem 4 in Chapter 3, Book III.) A function $f$ on $\Omega \subset \mathbb{R}^d$ is convex if the restriction of $f$ to any line segment in $\Omega$ is convex.

(a) Suppose $f$ is continuous on $(a, b)$. Then it is convex if and only if the distribution $\frac{d^2f}{dx^2}$ is positive.

(b) If $f$ is continuous on $\Omega \subset \mathbb{R}^d$, it is convex if and only if for each $\xi = (\xi_1, \ldots, \xi_d) \subset \mathbb{R}^d$ the distribution $\sum_{1 \leq i,j, \leq d} \xi_j \xi_j \frac{\partial^2 f}{\partial x_i \partial x_j}$ is positive.

[Hint: For (a), let $\varphi \in \mathcal{D}$, $\varphi \geq 0$, $\int \varphi \, dx = 1$ and set $\varphi_\epsilon(x) = \epsilon^{-1}\varphi(x/\epsilon)$. Consider $f_\epsilon = f * \varphi_\epsilon$.]

**27.** Every distribution $F$ of compact support in $\mathbb{R}^d$ is of **finite order** in the following sense: for each such $F$, there exists an integer $M$ and continuous functions $F_\alpha$ of compact support, so that

$$F = \sum_{|\alpha| \leq M} \partial_x^\alpha F_\alpha.$$

Moreover if $F$ is supported in $C$, then for every $\epsilon > 0$ we may take $F_\alpha$ to be supported in an $\epsilon$-neighborhood of $C$. Prove this by carrying out the following three steps.

(a) Pick $N$ so that $|F(\varphi)| \leq c\|\varphi\|_N$, for all $\varphi \in \mathcal{S}$, and choose $M_0$ so that $2M_0 > d + N$. Let $Q$ be the inverse Fourier transform of $1/(1 + 4\pi^2|\xi|^2)^{M_0}$, and observe that $Q$ is a fundamental solution of $(1 - \triangle)^{M_0}$, and $Q$ is of class $C^N$.

(b) For each $\epsilon$, construct $Q_\epsilon$ corresponding to $Q$, so that $(1 - \triangle)^{M_0} Q_\epsilon = \delta + r_\epsilon$, where $Q_\epsilon$ is supported in the $\epsilon$-neighborhood of the origin (as in Corollary 2.13). Prove that $F * Q_\epsilon$ is a continuous function, using the fact that $|F(\varphi)| \leq c\|\varphi\|_N$.

(c) Hence $F = (1 - \triangle)^{M_0}(Q_\epsilon * F) - F * r_\epsilon$, and the result is proved with $M = 2M_0$.

**28.** One can characterize tempered distributions $F$ whose Fourier transforms have compact support.

We already know by Proposition 1.6 that such an $F$ must in fact be a function $f$ that is $C^\infty$ and slowly increasing. A precise characterization when $d = 1$ is given in the statement below.

The Fourier transform of a tempered distribution $F$ is supported in the interval $[-M, M]$ if and only $F$ equals a function $f$ that is $C^\infty$, slowly increasing, and having an analytic extension to the complex plane as an entire function of exponential type $2\pi M$; that is, for every $\epsilon > 0$, $|f(z)| \leq A_\epsilon e^{2\pi(M+\epsilon)|z|}$, where $z = x + iy$.

(An analogous assertion holds in higher dimensions.)

[Hint: Assume $\hat{F}$ is supported in $[-M, M]$. Using Exercise 27 allows us to write $\hat{F} = \sum_{|\alpha| N} \partial_x^\alpha(g_\alpha)$, where $g_\alpha$ are continuous and supported in $[-M - \epsilon, M + \epsilon]$, and thus reduce to the case when $\hat{F}$ is a continuous function.

To prove the converse, consider $f_\delta = f\gamma_\delta$ where $\gamma_\delta(x) = \frac{1}{\delta}\int e^{-2\pi i x\xi}\eta(\xi/\delta)\,d\xi$ with $\eta \in C^\infty$, supported in $|\xi| \leq 1$ and such that $\int \eta = 1$. Then $\gamma_\delta(z)$ is of exponential type $2\pi\delta$ and is rapidly decreasing on the real axis. Thus apply the simpler version of the result given in Theorem 3.3, Chapter 4 in Book II to the function $f_\delta$, and let $\delta \to 0$.]

**29.** In this exercise, we consider the $L^2$ Sobolev spaces.

The space $L^2_m$ consists of the functions $f \in L^2(\mathbb{R}^d)$ whose derivatives $\partial_x^\alpha f$ taken in the sense of distributions, are in $L^2(\mathbb{R}^d)$ for all $|\alpha| \le m$. This space is sometimes denoted by $H_m(\mathbb{R}^d)$. Note that this is the special case for $p = 2$ of the Sobolev space given as an example in Section 3 of Chapter 1. However, here we use a slightly different (but equivalent) norm, which makes $L^2_m$ into a Hilbert space.

On $L^2_m$ we define the inner product

$$(f, g)_m = \sum_{|\alpha| \le m} (\partial_x^\alpha f, \partial_x^\alpha g)_0,$$

with $(f, g)_0 = \int_{\mathbb{R}^d} f(x)\overline{g(x)}\, dx$. Then, $L^2_m$ with the norm $\|f\|_{L^2_m} = (f, f)_m^{1/2}$ is a Hilbert space.

(a) Verify that $f \in L^2_m$ if and only if $\hat{f}(\xi)(1 + |\xi|)^m \in L^2$, and that the norms $\|f\|_{L^2_m}$ and $\|\hat{f}(\xi)(1 + |\xi|)^m\|_{L^2}$ are equivalent.

(b) If $m > d/2$, then $f$ can be corrected on a set of measure zero, so that $f$ becomes continuous and is in fact in $C^k$, for $k < m - d/2$. This is a version of the Sobolev embedding theorem.

[Hint: $\xi^\alpha \hat{f} \in L^1(\mathbb{R}^d)$ if $|\alpha| < m - d/2$.]

**30.** The following observation is useful in connection with the $L^2$ theory of Calderón-Zygmund distributions on $\mathbb{R}^d$.

(a) The Fourier transform of the distribution $\left[\frac{1}{|x|^d}\right]$ equals $c_1 \log|\xi| + c_2$, with $c_1 \neq 0$.

(b) Prove the following consequence of (a). Suppose $k$ is a homogeneous function of degree $-d$ that is $C^\infty$ away from the origin and with

$$\int_{|x|=1} k(x)\, d\sigma(x) \neq 0.$$

If $K$ is any distribution that agrees with $k$ away from the origin, then the Fourier transform of $K$ is not a bounded function. Another way of stating this is that the operator $T$, defined by $T(\varphi) = K * \varphi$ initially defined for $\varphi \in \mathcal{D}$, does not extend to a bounded operator on $L^2(\mathbb{R}^d)$.

**31.** Suppose $k$ is a $C^\infty$ function homogenenous of degree $-d$, not identically equal to zero, and

$$\int_{|x|=1} k(x)\, d\sigma(x) = 0.$$

If $K$ is the principal value distribution defined by $k$, that is, $K = \mathrm{pv}(k)$, then $K$ is a Calderón-Zygmund distribution but the operator $T$ given by $Tf = f * K$ is not bounded on $L^1$ or $L^\infty$.

The special case of the Hilbert transform is in Exercise 7, Chapter 2.

[Hint: If $\varphi \in \mathcal{D}$, then $T\varphi(x) = ck(x) + O(|x|^{-d-1})$ as $|x| \to \infty$, where $c = \int \varphi$.]

**32.** The cancelation condition (21) for the Calderón-Zygmund distributions for some $n > 1$ implies the condition for $n = 1$. Show this by first proving the following fact: Whenever $K$ satisfies (20) and (21) for some $n \geq 1$, then for every $1 \leq j \leq d$, the distribution $x_j \cdot K$ equals the locally integrable function $x_j k$.

[Hint: The distribution $x_j K - x_j k$ is supported at the origin. Then use Theorem 1.7 to test $x_j K - x_j k$ against $\varphi_r$ as $r \to 0$ for suitable $\varphi$, to conclude that this difference vanishes. Next, write any $C^{(1)}$-normalized bump function as $\varphi(x) = \eta(x) + \sum_{j=1}^{d} x_j \varphi_j(x)$ where $\eta$ and the $\varphi_j$ are multiples of $C^{(n)}$ and $C^{(0)}$-normalized bump functions respectively, and use the above fact.]

**33.** Suppose $k$ is a $C^\infty$ function in $\mathbb{R}^d - \{0\}$, that satisfies the differential inequalities (20). Then there is a Calderón-Zygmund distribution $K$ which has $k$ as its associated function if and only if $\sup_{0<a<b} \left| \int_{a<|x|<b} k(x)\,dx \right| < \infty$.

[Hint: In one direction, note that $|K(\eta_b - \eta_a)| \leq 2A$, where $\eta(x) = 1$ if $|x| \leq 1/2$, and $\eta(x) = 0$ if $|x| \geq 1$, with $\eta \in C^\infty$. In the other direction, define

$$K(\varphi) = \int_{|x|\leq 1} k(x)(\varphi(x) - \varphi(0))\,dx + \int_{|x|\geq 1} k(x)\varphi(x)\,dx$$

and verify that the conditions (20) and (21) hold for $K$.]

**34.** Suppose $H$ is a Calderón-Zygmund distribution and $\eta$ belongs to $\mathcal{S}$. Verify that $\eta K$ is a Calderón-Zygmund distribution.

## Chapter 4. Applications of the Baire Category Theorem

**1.** Below are some examples of generic sets and sets of the first category.

(a) Let $\{x_j\}_{j=1}^{\infty}$ denote an enumeration of the rational numbers in $\mathbb{R}$, and consider the sets

$$U_n = \bigcup_{j=1}^{\infty}(x_j - \frac{1}{n2^j},\ x_j + \frac{1}{n2^j}), \qquad \text{and} \qquad U = \bigcap_{n=1}^{\infty} U_n.$$

Show that $U$ is generic but has Lebesgue measure zero.

(b) Use a Cantor-like set (as described, for example, in Exercise 4, Chapter 1 of Book III) to give an example of a subset of the first category that has full Lebesgue measure in $[0, 1]$. Note that automatically this subset will be uncountable and dense. Also, its complement is generic and has measure zero, giving an alternative to the set $U$ in (a).

**2.** Suppose $F$ is a closed subset and $\mathcal{O}$ an open subset of a complete metric space.

(a) Show that $F$ is of the first category if and only if $F$ has empty interior.

(b) Show that $\mathcal{O}$ is of the first category if and only if $\mathcal{O}$ is empty.

(c) Consequently, prove that $F$ is generic if and only if $F = X$; and $\mathcal{O}$ is generic if and only if $\mathcal{O}^c$ contains no interior.

[Hint: For (a), argue by contradiction, assuming that a closed ball $\overline{B}$ is contained in $F$. Apply the category theorem to the complete metric space $\overline{B}$.]

**3.** Show that the conclusion of the Baire category theorem continues to hold if $X_0$ is a metric space that arises as an open subset of a complete metric space $X$.

[Hint: Apply the Baire category theorem to the closure of $X_0$ in $X$.]

**4.** Prove that every continuous function on $[0, 1]$ can be approximated uniformly by continuous nowhere differentiable functions. Do so by either:

(a) using Theorem 1.5.

(b) using only the fact that a continuous nowhere differentiable function exists.

**5.** Let $X$ be a complete metric space. We recall that a set is a $G_\delta$ in $X$ if it is a countable intersection of open sets. Also, a set is an $F_\sigma$ in $X$ if it is a countable union of closed sets.

(a) Show that a dense $G_\delta$ is generic.

(b) Hence a countable dense set is an $F_\sigma$, but not a $G_\delta$.

(c) Prove the following partial converse to (a). If $E$ is a generic set, then there exists $E_0 \subset E$ with $E_0$ a dense $G_\delta$.

**6.** The function

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1/q & \text{if } x = p/q \text{ is rational and expressed in lowest form} \end{cases}$$

is continuous precisely at the irrationals. In contrast to this, prove that there is no function on $\mathbb{R}$ that is continuous precisely at the rationals.

[Hint: Show that the set of points where a function is continuous is a $G_\delta$ (see the proof of Theorem 1.3), and apply Exercise 5.]

**7.** Let $E$ be a subset of $[0,1]$, and let $I$ be any closed non-trivial interval in $[0,1]$.

(a) Suppose $E$ is of the first category in $[0,1]$. Show that for every $I$, the set $E \cap I$ is of the first category in $I$.

(b) Suppose $E$ is generic in $[0,1]$. Show that for every $I$, the set $E \cap I$ is generic in $I$.

(c) Construct a set $E$ in $[0,1]$ so that for all $I$, the set $E \cap I$ is neither of the first category nor generic in $I$.

[Hint: Consider the Cantor set in $[0,1]$; then in each open interval of its complement place a scaled copy of the Cantor set; continue this process indefinitely. For a related measure theoretic result, see Exercise 36 in Chapter 1 of Book III.]

**8.** A **Hamel basis** for a vector space $X$ is a collection $\mathcal{H}$ of vectors in $X$, such that any $x \in X$ can be written as a unique *finite* linear combination of elements in $\mathcal{H}$.

Prove that a Banach space cannot have a countable Hamel basis.

[Hint: Show that otherwise the Banach space would be of the first category in itself.]

**9.** Consider $L^p([0,1])$ with Lebesgue measure. Note that if $f \in L^p$ with $p > 1$, then $f \in L^1$. Show that the set of $f \in L^1$ so that $f \notin L^p$, is generic.

A more general result can be found in Problem 1.

[Hint: Consider the set $E_N = \{f \in L^1 : \int_I |f| \leq Nm(I)^{1-1/p}$ for all intervals $I\}$. Note that each $E_N$ is closed and that $L^p \subset \bigcup_N E_N$. Finally, show that $E_N$ is nowhere dense by considering $f_0 + \epsilon g$ where $g(x) = x^{-(1-\delta)}$ with $0 < \delta < 1 - 1/p$.]

**10.** Consider $\Lambda^\alpha(\mathbb{R})$, with $0 < \alpha < 1$. Show that the set of nowhere differentiable functions is a generic set in $\Lambda^\alpha(\mathbb{R})$.

Note however that functions corresponding to the case $\alpha = 1$, that is, Lipschitz functions, are almost everywhere differentiable. (See Exercise 32 in Chapter 3 of Book III.)

**11.** Consider the Banach space $X = C([0,1])$ over the reals, with the sup-norm on $X$. Let $\mathcal{M}$ be the collection of functions that are not monotonic (increasing or decreasing) in any interval $[a,b]$, where $0 \leq a < b \leq 1$. Prove that $\mathcal{M}$ is generic in $X$.

[Hint: Let $\mathcal{M}_{[a,b]}$ denote the subset of $X$ consisting of functions that are not monotonic in $[a,b]$. Then $\mathcal{M}_{[a,b]}$ is dense in $X$, while $\mathcal{M}^c_{[a,b]}$ is closed.]

**12.** Suppose $X$, $Y$ and $Z$ are Banach spaces, and $T : X \times Y \to Z$ is a mapping such that:

(i) For each $x \in X$, the mapping $y \mapsto T(x, y)$ is linear and continuous on $Y$.

(ii) For each $y \in Y$, the mapping $x \mapsto T(x, y)$ is linear and continuous on $X$.

Prove that $T$ is (jointly) continuous on $X \times Y$, and in fact,

$$\|T(x, y)\|_Z \leq C \, \|x\|_X \, \|y\|_Y$$

for some $C > 0$ and all $x \in X$ and $y \in Y$.

**13.** Let $(X, \mathcal{F}, \mu)$ be a measure space, and let $\{f_n\}$ a sequence of functions in $L^p(X, \mu)$. We know from Exercise 12 in Chapter 1, that if $1 < p < \infty$, and $\sup_n \|f_n\|_{L^p} < \infty$, then some subsequence of $\{f_n\}$ converges weakly in $L^p$. In other words, there exist a subsequence $\{f_{n_k}\}$ of $\{f_n\}$, and an $f \in L^p$, so that if $q$ denotes the conjugate exponent of $p$, that is $1/p + 1/q = 1$, then

$$\int_X f_{n_k}(x)g(x)\ d\mu(x) \to \int_X f(x)g(x)\ d\mu(x) \qquad \text{for every } g \in L^q.$$

More generally, we say that a sequence $\{f_n\}$ in $L^p$ is **weakly bounded** if

$$\sup_n \left| \int_X f_n(x)g(x)\ d\mu(x) \right| < \infty \qquad \text{for all } g \in L^q.$$

Prove that if $1 < p < \infty$, and $\{f_n\}$ is a sequence of functions in $L^p$ that is weakly bounded, then

$$\sup_n \|f_n\|_{L^p} < \infty.$$

In particular this holds if $\{f_n\}$ converges weakly in $L^p$.

[Hint: Apply the uniform boundedness principle to $\ell_n(g) = \int_X f_n(x)g(x)\ d\mu(x)$.]

**14.** Suppose $X$ is a complete metric space with respect to a metric $d$, and $T : X \to X$ a continuous function. An element $x^*$ in $X$ is **universal** for $T$ if the orbit set $\{T^n(x^*)\}_{n=1}^{\infty}$ is dense in $X$. Here $T^n = T \circ T \circ \cdots \circ T$ denotes $n$ compositions of $T$.

Show that the set of universal elements for $T$ in $X$ is either empty or generic.

[Hint: Suppose $x^*$ is universal for $T$, let $\{x_j\}$ be a dense set of elements in $X$, and let $F_{j,k,N} = \{x \in X : \ d(T^n x, y_j) < 1/k \ \text{ for some } n \geq N\}$. Show that $F_{j,k,N}$ is open and dense.]

**15.** Let $\overline{B}$ denote the closure of the unit ball in $\mathbb{R}^d$, and consider the metric space $\mathcal{C}$ of compact subsets of $\overline{B}$ with the Hausdorff distance. (See Section 5.) Show that the following two collections are generic.

(a) The subsets of Lebesgue measure zero.

(b) The subsets that are nowhere dense.

[Hint: For (a) show that the collection of sets $C$ so that $m(C) < 1/n$ is open and dense. In fact for such a set, $C^c \supset \bigcup_{j=1}^{M} Q_j$, where $Q_j$ are disjoint open cubes so that $\sum |Q_j| > 1 - 1/n$. Now shrink the $Q_j$. For (b) fix an open set $\mathcal{O}$ and show that the collection $\mathcal{C}_\mathcal{O}$ of sets in $\mathcal{C}$ that contain $\mathcal{O}$ is closed and nowhere dense.]

## Chapter 5. Rudiments of Probability Theory

**1.** Consider $S_N(x) = \sum_{n=1}^{N} r_n(x)$, with $N$ odd.

(a) Calculate $m(\{x : \ S_N(x) = k\})$, and show that as $k$ varies over the integers, the maximum is attained at $k = -1$ and $k = 1$.

(b) Adapt the proof when $N$ is even to show that for odd $N$,

$$m(\{x : \ a < \frac{S_N(x)}{N^{1/2}} < b\}) \to \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2/2}\, dt, \qquad \text{as } N \to \infty.$$

**2.** Find three functions $f_1$, $f_2$, and $f_3$, so that any pair are mutually independent, but the three are not.

[Hint: Let $f_1 = r_1$, $f_2 = r_2$, and express $f_3$ in terms of $r_1$ and $r_2$.]

**3.** The collection $\{r_n\}$ of mutually independent functions on $[0, 1]$ cannot be much enlarged and still remain mutually independent. In fact, prove that if we adjoin a function $f$ to the collection $\{r_n\}$, then the resulting collection is also mutually independent only when $f$ is constant.

[Hint: See also Exercise 16.]

**4.** Suppose $\mu$ and $\nu$ are two finite measures on a space $X$ that agree on a collection of sets $\mathcal{C}$. If $\mathcal{C}$ contains $X$ and is closed under finite intersections, then show that $\mu = \nu$ on the $\sigma$-algebra generated by $\mathcal{C}$.

[Hint: The equality $\mu = \nu$ holds on finite unions of sets in $\mathcal{C}$ because

$$\mu(\bigcup_{j=1}^{k} C_j) = \sum_{j=1}^{k} \mu(C_j) - \sum_{i<j} \mu(C_i \cap C_j) + \\ + \sum_{i<j<\ell} \mu(C_i \cap C_j \cap C_\ell) + \cdots + (-1)^{k-1} \mu(\bigcap_{j=1}^{k} C_j).]$$

**5.** Prove that the $\mathbb{R}^d$-valued functions $f_1, \ldots, f_n$ are mutually independent if and only if their joint distribution measure equals the product of the individual distribution measures:

$$\mu_{f_1,\ldots,f_n} = \mu_{f_1} \times \cdots \times \mu_{f_n} \qquad \text{as measures on } \mathbb{R}^{nd} = \mathbb{R}^d \times \cdots \times \mathbb{R}^d.$$

[Hint: Check the equality on cylinder sets in $\mathbb{R}^{nd}$ and use the previous exercise.]

**6.** Suppose $\{f_n\}$ is a sequence of mutually independent functions on the probability space $(X, m)$. Prove that there exists a probability space $(X', m')$, with $X'$ an infinite product, $X' = \prod_{n=1}^{\infty} X_n$, $(X_n, m_n)$ probability spaces, and $m'$ the product measure of the $m_n$, so that the following holds: there are functions $\{g_n\}$ on $X'$ so that $\{f_n\}$ and $\{g_n\}$ have the same joint distributions, but each function $g_n$ depends only on the $n^{\text{th}}$ coordinate of $X'$.

[Hint: Take $(X_n, m_m) = (X, m)$ for each $n$ and define $g_n$ in terms of the $f_n$ accordingly, and use the previous exercise.]

**7.** Show that if $\mathcal{B}_0, \ldots, \mathcal{B}_n$ are mutually independent algebras, then for each $k < n$, the algebras $\bigvee_{j=0}^{k} \mathcal{B}_j$ and $\mathcal{B}_n$ are mutually independent.

Prove this by noting the following. First, use induction to show that if $B_j \in \mathcal{B}_j$, then $B_0 \cap \cdots \cap B_k$ is independent of $B_n$. Now fix $B \in \mathcal{B}_n$, and consider the two finite measures $\mu(E) = m(E \cap B)$ and $\nu(E) = m(E)m(B)$, and the collection $\mathcal{C}$ of sets that are of the form $E = B_0 \cap \cdots \cap B_k$, where $B_j \in \mathcal{B}_j$. Then apply Exercise 4.

**8.** Verify the following further facts about probability distribution measures.

(a) Suppose $f = (f_1, \ldots, f_k)$ with each $f_j$ an $\mathbb{R}^d$-valued function. Let $\mu$ be the probability distribution measure of $f$, and let $L$ be a linear transformation of $\mathbb{R}^{dk}$ to itself. Suppose that $\mu$ is the probability distribution measure of $f$. Then the distribution measure of $L(f)$ is $\mu_L$, where $\mu_L(A) = \mu(L^{-1}A)$ for every Borel set $A \subset \mathbb{R}^{dk}$.

(b) Suppose the distribution measure of $f_j$ is Gaussian with covariance matrix $\sigma_j^2 I$, $1 \leq j \leq k$. Assume also that the $\{f_j\}$ are mutually independent. Then the distribution measure of $c_1 f_1 + \cdots + c_k f_k$ is Gaussian with covariance matrix $(c_1^2\sigma_1^2 + \cdots + c_k^2\sigma_k^2)I$.

[Hint: For (b), compute the Fourier transforms (the characteristic functions) of the measures in question.]

**9.** Consider the space $L^2(\Omega, \mathbb{R}^d)$ of square-integrable $\mathbb{R}^d$-valued functions on the probability space $(\Omega, P)$. A closed subspace $\mathcal{G}$ of this space is called a **Gaussian subspace** if it is spanned by a sequence $\{f_n\}$ of mutually independent functions, each having a Gaussian distribution measure with mean zero, and covariance $\{\sigma_n^2 I\}$.

Prove that if $F_1, F_2, \ldots, F_k$ are mutually orthogonal elements of $\mathcal{G}$, then they are mutually independent. Note that the converse of this is immediate.

[Hint: Consider the case when $\mathcal{G}$ is finite-dimensional, and $\mathcal{G}$ is spanned by $f_1, \ldots, f_N$. One may suppose, after multiplication by appropriate scalars, that the $f_j$ and $F_j$ each have $L^2$ norm equal to 1. So there exists an orthogonal linear transformation $L$ so that $L(f_j) = F_j$. Then apply Exercises 5 and 8.]

**10.** Consider the following two types of convergence of a sequence $\{f_n\}$ to a limit $f$ on a probability space:

(i) $f_n \to f$ almost everywhere,

(ii) $f_n \to f$ in terms of weak convergence of their distribution measures.

Prove that (i) implies (ii), but that this implication cannot be reversed.

[Hint: Recall that if $\varphi$ is continuous and bounded, then $\int \varphi(f)\, d\mu = \int \varphi\, d\mu_f$, where $\mu_f$ is the distribution measure of $f$, and apply the dominated convergence theorem.]

**11.** On $[0,1]$ with Lebesgue measure, construct a function $f$ whose distribution measure is normal.

[Hint: Consider the "error function" $\mathrm{Erf}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt$ and its inverse function.]

**12.** Prove the identity (6), which says that if $G$ is a non-negative continuous function on $\mathbb{R}$ (or continuous and bounded), and $f$ is a real-valued measurable function (on a probability space $(X, m)$) with distribution measure $\mu = \mu_f$, then

$$\int_X G(f)(x)\, dm = \int_{\mathbb{R}} G(t)\, d\mu(t).$$

[Hint: Note that if $f$ is bounded, then $\sum_k G(k/n)\, m(\{k/n < f < (k+1)/n\})$ converges to both integrals as $n \to \infty$.]

**13.** The Rademacher sequence $\{r_n\}$ is far from complete on $L^2([0,1])$. In fact it cannot be completed by adjoining any finite collection of functions. Prove this in two ways.

(a) By considering the functions $\{r_n r_m\}$ for $n < m$.

(b) By using the $L^p$ inequality of Lemma 1.8.

See also Exercise 16.

**14.** Consider the power series

$$\sum_{n=1}^{\infty} \pm a_n z^n = \sum_{n=1}^{\infty} r_n(t) a_n z^n = F(z,t),$$

where $\sum |a_n|^2 = \infty$ and $\limsup |a_n|^{1/n} \leq 1$.

Show that for almost every $t$, the function $F(z,t)$ cannot be analytically continued outside the unit disc.

[Hint: Argue as in Theorem 1.7 part (b) using Abel summation rather than Cesàro summation.]

**15.** Show that the $L^2([0,1])$ span of $\{r_n\}$ can be characterized as the subspace of $L^2$ consisting of those $f$ for which

$$\mathbb{E}_N(f) = S_N(f), \qquad \text{for all } N,$$

where $\mathbb{E}_N$ are the conditional expectations corresponding to the dyadic intervals of length $2^{-N}$.

**16.** A natural completion of the collection of Rademacher functions are the **Walsh-Paley functions**. One defines this collection on $[0,1]$, denoted by $\{w_n\}$, in the following way.

First one sets $w_0(t) = 1$, $w_1(t) = r_1(t)$, $w_2(t) = r_2(t)$ and $w_3(t) = r_1(t)r_2(t)$. More generally, if $n \geq 1$, $n = 2^{k_1} + 2^{k_2} + \cdots + 2^{k_\ell}$, where $0 \leq k_1 < \cdots < k_\ell$, then one defines

$$w_n(t) = \prod_{j=1}^{\ell} r_{k_j+1}(t).$$

In particular $w_{2^{k-1}} = r_k$.

(a) Prove that $\{w_n\}_{n=0}^{\infty}$ is a complete orthonormal system on $L^2([0,1])$.

(b) Verify the following additional interesting property of the Walsh-Paley functions: they are the continuous characters of the compact abelian group $\mathbb{Z}_2^\infty$ (thought of as the product of the two-point abelian groups $\mathbb{Z}_2$).

[Hint: Equip the group $\mathbb{Z}_2^\infty$ with the addition $x + y$ defined by $(x+y)_j = x_j + y_j \mod 2$ if $x = (x_j)$ and $y = (y_j)$. Then $r_k(x+y) = r_k(x)r_k(y)$.

Consider also the "Dirichlet kernel" $K_N(t) = \sum_{k=0}^{N-1} w_k(t)$, and show that if $N = 2^n$, then $K_N(t) = \prod_{j=1}^{n}(1 + r_j(t))$, hence $K_{2^n}(t) = 2^n$ if $0 \leq t \leq 2^{-n}$ and 0 otherwise. As a result, using the convolution $\int f(y)K_N(x+y)\,dy$, note that if $f \sim \sum a_k w_k$, then $\sum_{k<2^n} a_k w_k = \mathbb{E}_n(f)$, where $\mathbb{E}_n$ was defined in Section 1.6. See also Problem $2^*$.]

**17.** The inequality in Lemma 1.8 may be strengthened as follows. Let $F(t) = \sum_{n=1}^{\infty} a_n r_n(t)$, with $a_n$ real and $\sum_{n=1}^{\infty} a_n^2 = 1$. Then

(a) $\int_0^1 e^{\mu|F(t)|}\, dt \leq 2e^{\mu^2}$, for all $0 \leq \mu$.

(b) As a result, for some $c > 0$, $\int_0^1 e^{c|F(t)|^2}\, dt < \infty$.

[Hint: Part (a) implies that $m(\{t : |F(t)| > \alpha\}) \leq 2e^{\mu^2 - \mu\alpha}$. Choose $\mu = \alpha/2$, and obtain (b) with $c < 1/4$.]

**18.** Prove that there exists an $f$ in $L^p([0, 2\pi])$, for all $p < \infty$, $f \sim \sum_{-\infty}^{\infty} c_n e^{in\theta}$, so that $\sum |c_n|^q = \infty$, for all $q < 2$. Hence the Hausdorff-Young inequality in Section 2.1 of Chapter 2 fails for $p > 2$.

[Hint: Use Theorem 1.7.]

**19.** Suppose $F(t) = \sum_{n=1}^{\infty} a_n r_n(t)$ with $\sum a_n^2 < \infty$.

(a) Prove directly that there exists a constant $A$ so that

$$\|F\|_{L^4} \leq A\|F\|_{L^2}.$$

(b) Show as a result that there is a constant $A'$ so that $\|F\|_{L^2} \leq A'\|F\|_{L^1}$.

(c) Conclude that $\|F\|_{L^p} \leq A_p\|F\|_{L^1}$, for $1 \leq p < \infty$.

[Hint: For (a) write out $\int_0^1 F^4(t)\,dt$ as a sum and use the orthogonality of $r_n(t)r_m(t)$. For (b) use Hölder's inequality. For (c) use Lemma 1.8.]

**20.** Suppose $\{A_n\}$ is a sequence of subsets of the probability space $X$.

(a) If $\sum m(A_n) < \infty$, then $m(\limsup_{n\to\infty} A_n) = 0$, where $\limsup_{n\to\infty} A_n$ is defined as $\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$.

(b) However if $\sum m(A_n) = \infty$, and the sets $\{A_n\}$ are mutually independent, then $m(\limsup_{n\to\infty} A_n) = 1$.

This dichotomy is often referred to as the Borel-Cantelli lemma. (See also Book III.)
[Hint: In the case (b), $m\left(\bigcap_{k=r}^{n} A_k^c\right) = \prod_{k=r}^{n}(1 - m(A_k))$.]

**21.** Except for a countable set (the dyadic rationals) it is possible to assign a unique dyadic expansion to each real number $\alpha$ in $[0,1]$, that is,

$$\alpha = \sum_{j=1}^{\infty} \frac{x_j}{2^j} \qquad \text{with } x_j = 0 \text{ or } 1.$$

Given such a number $\alpha$ let $\#_N(\alpha)$ denote the number of 1's that appear among the first $N$ terms in the dyadic expansion of $\alpha$. We say that $\alpha$ is **normal**, if its dyadic expansion contains a density of 1's equal to the density of 0's, that is,

$$\lim_{N\to\infty} \frac{\#_N(\alpha)}{N} = 1/2.$$

(a) Prove that (with respect to the Lebesgue measure) almost every number in $[0,1]$ is normal.

(b) More generally, given an integer $q \geq 2$ consider the $q$-expansion of a real number $\alpha$ in $[0,1]$,

$$\alpha = \sum_{j=1}^{\infty} \frac{x_j}{q^j} \qquad \text{with } x_j = 0, 1, \ldots, q-1. \tag{21}$$

Again, ignoring a countable subset, this expansion is unique. For a given real number $\alpha$ and for each $0 \leq p \leq q-1$, define $\#_{p,N}(\alpha)$ to equal the number of $j$'s with $0 \leq j \leq N$ for which $x_j = p$ in the $q$-expansion of $\alpha$. A number that satisfies

$$\lim_{N\to\infty} \frac{\#_{p,N}(\alpha)}{N} = 1/q,$$

for every $0 \leq p \leq q-1$ is said to be **normal** to base $q$.

Show that almost all real numbers in $[0,1]$ satisfy this property.

[Hint: Consider the infinite product $\prod \mathbb{Z}_q$ with each factor given the uniform measure. Under (21) the product measure corresponds to the Lebesgue measure on $[0,1]$. Now apply the law of large numbers as in Theorem 2.1.]

**22.** A sequence $\{f_n\}_{n=0}^{\infty}$ of functions on $X$ is called a (discrete) **stationary process** if for every $N$ the joint probability distribution of $f_r, f_{r+1}, \ldots, f_{r+N}$ is independent of $r$.

Consider the probability space $Y$ constructed as in the proof of Theorem 2.1. Show that whenever the sequence $\{f_n\}$ is a stationary process, then it has the same joint distribution as the sequence $\{g_0(\tau^n(y))\}$, where $g_0$ is a suitable function on $Y$ and $\tau$ is the shift. Hence the ergodic theorem is equally applicable in this more general situation.

**23.** Prove that the conditions in Theorem 2.1 are sharp in the following sense. If $\{f_n\}_{n=0}^{\infty}$ are mutually independent and identically distributed, but $\int_X |f_0(x)|\, dm = \infty$, then for almost every $x$, the averages $\frac{1}{N}\sum_{n=0}^{N-1} f_n(x)$ fail to converge to a limit as $N \to \infty$.

[Hint: Let $A_n = \{x : \ |f_n(x)| > n\}$. The sets $A_n$ are independent. However, $\sum_{n=0}^{\infty} m(A_n) = \sum_{n=0}^{\infty} m(\{x : \ |f_0(x)| > n\}) \approx \int_X |f_0(x)|\, dm = \infty$. Then use Exercise 20.]

**24.** The following are examples of conditional expectations.

(a) Suppose $X = \bigcup A_n$ is a finite (or countable) partition of $X$, with $m(A_n) > 0$ whenever $A_n$ is non-empty. Let $\mathcal{A}$ be the algebra generated by the sets $\{A_n\}$. Then $\mathbb{E}_{\mathcal{A}}(f)(x) = \frac{1}{m(A_n)} \int_{A_n} f \, dm$ whenever $x \in A_n$.

(b) Let $X = X_1 \times X_2$, with the measure $m$ on $X$ being the product of the measures $m_i$ on $X_i$. Let $\mathcal{A} = \{A \times X_2\}$, where $A$ ranges over arbitrary measurable sets of $X_1$. Then $\mathbb{E}_{\mathcal{A}}(f)(x_1, x_2) = \int_{X_2} f(x_1, y) \, dm_2(y)$.

**25.** In the following four exercises $\{s_n\}$ will denote a martingale sequence corresponding to the increasing sequence of algebras $\mathcal{A}_n$ and their conditional expectations $\mathbb{E}_n$.

Suppose $s_n = \mathbb{E}_n(s_\infty)$ with $s_\infty \in L^2$. Then $\{s_n\}$ converges in $L^2$.

[Hint: Note that if $f_n = s_n - s_{n-1}$ then the $f_n$'s are mutually orthogonal and $s_n - s_0 = \sum_{k=1}^{n} f_k$.]

**26.** Prove the following.

(a) If $s_\infty \in L^p$, then $s_n = \mathbb{E}_n(s_\infty) \in L^p$, and $\|s_n\|_{L^p} \leq \|s_\infty\|_{L^p}$ for all $p$ with $1 \leq p \leq \infty$.

(b) Conversely, if $\{s_n\}$ is martingale and $\sup_n \|s_n\|_{L^p} < \infty$, then there exists $s_\infty \in L^p$, so that $s_n = \mathbb{E}_n(s_\infty)$, when $1 < p \leq \infty$.

(c) Show, however, that the conclusion in (b) may fail when $p = 1$.

[Hint: For (a) argue as in the proof of Lemma 2.5(a). For (b), use Lemma 2.5 and also the weak compactness of $L^p$, $p > 1$, as in Exercise 12 in Chapter 1. For (c), let $X = [0, 1]$ with Lebesgue measure, and consider $s_n(x) = 2^n$ for $0 \leq x \leq 2^{-n}$, $s_n(x) = 0$ otherwise.]

**27.** Suppose that $s_n = \mathbb{E}_n(s_\infty)$, with $s_\infty$ integrable on $X$.

(a) Show that $s_n$ converges in the $L^1$ norm as $n \to \infty$.

(b) Moreover $s_n \to s_\infty$ in $L^1$ if and only if $s_\infty$ is measurable with respect to the algebra $\mathcal{A}_\infty = \bigvee_{n=1}^{\infty} \mathcal{A}_n$.

[Hint: For (a) use Exercises 25 and 26 (a). Then $\lim s_n = \mathbb{E}_{\mathcal{A}_\infty}(s_\infty)$, and use the previous exercise.]

**28.** Suppose that $s_n = \mathbb{E}_n(s_\infty)$, and $s_\infty \in L^1$.

(a) Show that

$$m(\{x : \sup_n |s_n(x)| > \alpha\}) \leq \frac{1}{\alpha} \int_{|s_\infty(x)|>\alpha} |s_\infty(x)|\, dx.$$

(b) Prove as a result $\|\sup_n |s_n|\|_{L^p} \leq A_p \|s_\infty\|_{L^p}$ if $s_\infty \in L^p$ and $1 < p \leq \infty$.

[Hint: For (a), note that when $s_\infty \geq 0$ this is a consequence of (15). To deduce (b) adapt the argument in the proof of Theorem 4.1 in Chapter 2 for the maximal function $f^*$.]

**29.** The results for real-valued martingale sequences $\{s_n\}$ discussed in Section 2.2 go through if we assume that the $s_n$ take their values in $\mathbb{R}^d$. Verify in particular that the following consequences of identity (14) hold:

(a) $|s_k| \leq \mathbb{E}_k(|s_n|)$, if $k \leq n$.

(b) $m(\{x : \sup_n |s_n(x)| > \alpha\}) \leq \frac{1}{\alpha} \int_{|s_\infty(x)|>\alpha} |s_\infty(x)|\, dx$.

Here $|\cdot|$ denotes the Euclidean norm in $\mathbb{R}^d$.

[Hint: To prove (a), note that $(s_k, v) = \mathbb{E}_k((s_n, v))$, where $(\cdot, \cdot)$ is the inner product on $\mathbb{R}^d$, and $v$ is any fixed vector in $\mathbb{R}^d$. Then take the supremum over unit vectors $v$. The conclusion (b) is a consequence of (a) and part (a) of Exercise 28.]

**30.** The ideas regarding conditional expectations extend to spaces $(X, m)$ whose total measures are not necessarily finite. Consider the following example: $X = \mathbb{R}^d$, with $m$ the Lebesgue measure. For each $n \in \mathbb{Z}$, let $\mathcal{A}_n$ be the algebra generated by all dyadic cubes of side-length $2^{-n}$. The dyadic cubes are the open cubes, whose vertices are points of $2^{-n}\mathbb{Z}^d$, and have side-length $2^{-n}$. Clearly $\mathcal{A}_n \subset \mathcal{A}_{n+1}$ for all $n$. Let $f$ be integrable on $\mathbb{R}^d$, and set $\mathbb{E}_n(f) = \mathbb{E}_{\mathcal{A}_n}(f)$, with

$$\mathbb{E}_n(f)(x) = \frac{1}{m(Q)} \int_Q f \, dm$$

whenever $x \in Q$, with $Q$ a dyadic cube of length $2^{-n}$.

(a) Show that the maximal inequality in Theorem 2.10 extends to this case.

(b) If $f \geq 0$, then $\sup_{n\in\mathbb{Z}} \mathbb{E}_n(f)(x) \leq cf^*(x)$ for an appropriate constant $c$, with $f^*$ the Hardy-Littlewood maximal function discussed in Chapter 2.

(c) Show by example that the converse inequality $f^*(x) \leq c' \sup_{n\in\mathbb{Z}} \mathbb{E}_n(f)(x)$ fails. Prove however that a substitute result holds

$$m(\{x : f^*(x) > \alpha\}) \leq c_1 m(\{x : \sup_{n\in\mathbb{Z}} \mathbb{E}_n(f)(x) > c_2\alpha\})$$

for all $\alpha > 0$. Here $c_1$ and $c_2$ are appropriate constants.

**31.** Let $\{\mu_N\}_{N=1}^{\infty}$ and $\nu$ be probability measures on $\mathbb{R}^d$. Prove the following are equivalent as $N \to \infty$.

(a) $\hat{\mu}_N(\xi) \to \hat{\nu}(\xi)$, all $\xi \in \mathbb{R}^d$.

(b) $\mu_N \to \nu$ weakly.

(c) In $\mathbb{R}$, $\mu_N((a,b)) \to \nu((a,b))$ for all open intervals $(a,b)$, if we assume the measure $\nu$ is continuous.

(d) In $\mathbb{R}^d$, $\mu_N(\mathcal{O}) \to \nu(\mathcal{O})$ for all open sets $\mathcal{O}$, if we assume the measure $\nu$ is absolutely continuous with respect to Lebesgue measure.

[Hint: In $\mathbb{R}$, the equivalence of (a), (b) and (c) is implicit in the argument given in the proofs of Lemma 2.15 and Corollary 2.16. To show that (a) implies (d) in the case when $\mathcal{O}$ is an open cube, generalize the argument given in the text to $\mathbb{R}^d$. Then, prove that the analog of (d) holds for closed cubes. Finally, use the fact that any open set is an almost disjoint union of closed cubes. To show that (d) implies (b), approximate a continuous function $\varphi$ of compact support uniformly by step functions that are constant on cubes.]

**32.** The proof of Theorem 2.17 requires the following calculation. Suppose $\sigma$ is a strictly positive definite symmetric matrix with $\sigma^{-1}$ denoting its inverse. Let $\nu_{\sigma^2}$ be the measure on $\mathbb{R}^d$ with density equal to $\frac{1}{(2\pi)^{d/2}(\det \sigma)} e^{-\frac{|\sigma^{-1}(x)|^2}{2}}$, $x \in \mathbb{R}^d$. Then $\hat{\nu}_{\sigma^2}(\xi) = e^{-2\pi^2|\sigma(\xi)|^2}$.

[Hint: Verify this by making an orthogonal change of variables that puts $\sigma$ in a diagonal form. This reduces the $d$-dimensional integral in question to a product of corresponding 1-dimensional integrals.]

**33.** For the $d$-dimensional random walk $\{s_n(x)\}$ considered in Section 2.6, find the limit of the distribution measures of $s_n(x)/n^{1/2}$ as $n \to \infty$.

**34.** If $k$ is a lattice point in $\mathbb{Z}^d$ and $d = 1$ or 2, show that for almost every path, the random walk visits $k$ infinitely often, that is,

$$m(\{x : \ s_n(x) = k \quad \text{for infinitely many } n\} = 1.$$

[Hint: There exists $\ell_0$ so that $m(\{s_{\ell_0} = -k\}) > 0$. If the conclusion fails, then there exists $r_0$ so that $m(\{s_n \neq k, \text{ for all } n \geq r_0\}) > 0$. Then note that

$$\{s_n \neq 0, \text{ all } n \geq \ell_0 + r_0\} \supset \{s_{\ell_0} = -k\} \cap \{s_n - s_{\ell_0} \neq k, \text{all } n \geq \ell_0 + r_0\},$$

and that the sets on the right-hand side are independent.]

**35.** Prove that if $d \geq 3$, then the random walk $s_n$ satisfies $\lim_{n\to\infty} |s_n| = \infty$ almost everywhere.

[Hint: It is sufficient to prove that for any fixed $R > 0$ the set

$$B = \{x : \liminf_{n\to\infty} |s_n(x)| \leq R\}$$

has measure 0. To this end, for each lattice point $k$, define

$$B(k,\ell) = \{x : \ s_\ell(x) = k, \text{ and } s_n(x) = k \text{ for infinitely many } n\}.$$

Clearly, $B \subset \bigcup_{\ell,\ |k|\leq R} B(k,\ell)$. But $d \geq 3$, so $m(B(k,\ell)) = 0$).]

## Chapter 6. An Introduction to Brownian Motion

**1.** Show that if $t > 0$, then the distribution measure of $S_t^{(N)}$ converges weakly to the Gaussian $\nu_t$ with mean zero and variance $t$ as $N \to \infty$. More generally, if $t > s \geq 0$, then the distribution measure of $S_t^{(N)} - S_s^{(N)}$ converges weakly to the Gaussian $\nu_{t-s}$ with mean zero and covariance matrix $(t-s)I$.

[Hint: Using the notation in the remark following Theorem 2.17 in Chapter 5, and setting $f_k = \mathfrak{r}_k$, one has $S_t^{(N)} - S_{N,t} = \frac{(Nt-[Nt])}{N^{1/2}}\mathfrak{r}_{[Nt]+1}$.]

**2.** Let $(\mathcal{P}, d)$ be the metric space defined in Section 2. Verify:

(a) The space is complete.

(b) The space is separable.

[Hint: For (b), let $e_1, \ldots, e_d$ be a basis for $\mathbb{R}^d$, and consider the polynomials $p(t) = e_1 p_1(t) + \cdots + e_d p_d(t)$, where the $p_j$ have rational coefficients.]

**3.** Show that the metric space $(\mathcal{P}, d)$ is not $\sigma$-compact.

[Hint: Assume the contrary. Then the Baire category theorem implies that there exists a compact set that has a non-empty interior. As a result there exists an open ball whose closure is compact. However, consider for example the ball of radius 1 centered at 0, and a sequence of continuous piecewise linear functions $\{f_n\}$ with $f_n(0) = 1$, $f_n(x) = 0$, when $x \geq 1/n$.]

**4.** Suppose $X$ is a compact metric space. Show that:

(a) $X$ is separable.

(b) $C(X)$ is separable.

[Hint: For each $m$, find a finite collection $\mathcal{B}_m$ of open balls, each of radius $1/m$, so that the collection $\mathcal{B}_m$ covers $X$. For (a) take the centers of the balls in $\bigcup_{m=1}^{\infty} \mathcal{B}_m$. For (b), consider $\{\eta_k^{(m)}\}$ the partition of unity corresponding to the covering of $X$ by $\mathcal{B}_m$ (as given, for example, in Chapter 1). Show that the finite linear combinations of the $\eta_k^{(m)}$ with rational coefficients are dense in $C(X)$.]

**5.** Let $X$ be a metric space, $K \subset X$ a compact subset, and $f$ a continuous function on $K$. There there is a continuous function $F$ on $X$, so that

$$F|_K = f, \qquad \text{and} \qquad \sup_{x \in X} |F(x)| = \sup_{x \in K} |f(x)|.$$

[Hint: The argument given in Lemma 4.11, Chapter 5 of Book III for $X = \mathbb{R}^d$ can be copied over in this general setting.]

**6.** Suppose $K$ is a compact subset of $\mathcal{P}$. Show that for each $T > 0$, there exists a function $w_T(h)$, defined for $h \in (0, 1]$ with $w_T(h) \to 0$ as $h \to 0$ and such that

$$\sup_{\mathsf{p} \in K} \sup_{0 \le t \le T} |\mathsf{p}(t+h) - \mathsf{p}(t)| \le w_T(h), \qquad \text{for } h \in (0, 1].$$

[Hint: Fix $T > 0$ and $\epsilon > 0$. Each $\mathsf{p}$ is uniformly continuous on closed intervals, so there exists $\delta = \delta(\mathsf{p}) > 0$ so that $\sup_{0 \le t \le T} |\mathsf{p}(t+h) - \mathsf{p}(t)| \le \epsilon$ whenever $0 < h \le \delta$. Now use the fact that since $K$ is compact, the covering $K \subset \bigcup_{\mathsf{p}} \{\mathsf{p}' \in \mathcal{P} : d(\mathsf{p}', \mathsf{p}) < \epsilon\}$ has a finite subcover.]

**7.** Suppose $\mu_N \to \mu$ weakly. Show as a result:

(a) $\liminf_{N\to\infty} \mu_N(\mathcal{O}) \geq \mu(\mathcal{O})$ for any open set $\mathcal{O}$.

(b) $\lim_{N\to\infty} \mu_N(\mathcal{O}) = \mu(\mathcal{O})$, $\mathcal{O}$ an open set, if $\mu(\overline{\mathcal{O}} - \mathcal{O}) = 0$.

[Hint: $\mu(\mathcal{O}) = \sup_f \{\int f\, d\mu,$ where $0 \leq f \leq 1$ and $\text{supp}(f) \subset \mathcal{O}\}$.]

**8.** Given the Wiener measure $W$ in $\mathcal{P}$, we have a realization of Brownian motion (satisfying B-1, B-2 and B-3) with $B_t(\omega) = \mathsf{p}(t)$, $\Omega = \mathcal{P}$, and $P = W$. Conversely, suppose we start with $\{B_t\}$ satisfying B-1, B-2 and B-3. For any cylindrical set $C = \{\mathsf{p} \in \mathcal{P} : (\mathsf{p}(t_1), \ldots, \mathsf{p}(t_k)) \in A\}$ in $\mathcal{P}$, define $W^o(C) = P(\{\omega : (B_{t_1}(\omega), \ldots, B_{t_k}(\omega)) \in A\})$. Verify that $W^o$, initially defined on the cylindrical sets, extends to the Wiener measure on $\mathcal{P}$.

**9.** This exercise deals with the degree to which the Brownian motion process is uniquely determined by the properties B-1, B-2 and B-3.

Let us say that such a process is "strict" if in addition to the above it satisfies the following two conditions:

(i) $B_t(\omega_1) = B_t(\omega_2)$ for all $t$ implies $\omega_1 = \omega_2$.

(ii) The collection of measurable sets of $(\Omega, P)$ is exactly $\mathcal{A}_\infty$, which is the $\sigma$-algebra generated by the $\mathcal{A}_t$, with $t < \infty$.

Now given any Brownian motion process $B_t$ on $(\Omega, P)$ it induces a strict process $B_t^\#$ on $(\Omega^\#, P^\#)$ as follows: Let $\Omega^\#$ denote the collection of equivalence classes on $\Omega$ under the equivalence relation $\omega_1 \sim \omega_2$ if $B_t(\omega_1) = B_t(\omega_2)$ for all $t$. We also denote by $\{\omega\}$ the equivalence class to which $\omega$ belongs. On $\Omega^\#$ define $B_t^\#(\{\omega\}) = B_t(\omega)$, and $P^\#(\{A\}) = P(A)$ if $A \in \mathcal{A}_\infty$. Verify:

(a) $B_t^\#$ is a strict Brownian motion on $(\Omega^\#, P^\#)$.

(b) The process $(\mathcal{P}, W)$ constructed in Section 3 is a strict Brownian motion.

(c) If $(B_t^1, \Omega^1, P^1)$ and $(B_t^2, \Omega^2, P^2)$ are a pair of Brownian motion processes, then up to subsets of sets of measure zero, there is a bijection $\Phi : (\Omega^1)^\# \to (\Omega^2)^\#$, so that $(P^2)^\#(\Phi(A)) = (P^1)^\#(A)$ and $(B_t^2)^\#(\Phi(\omega)) = (B_t^1)^\#(\omega)$.

**10.** Prove the following version of Khinchin's inequality (Lemma 1.8 in the previous chapter). Suppose $\{f_n\}$ are identically distributed $\mathbb{R}^d$-valued functions that are bounded, have mean zero, and are mutually independent. Then for any $p < \infty$, we have

$$\| \sum a_n f_n \|_{L^p} \leq A_p \left( \sum |a_n|^2 \right)^{1/2}.$$

[Hint: One can reduce to the case $d = 1$. Assuming $\sum |a_n|^2 \leq 1$, write $\int e^{\sum a_n f_n} = \prod_n \int e^{a_n f_n}$ and use $e^u = 1 + u + O(u^2)$ if $|u| \leq M$. The result is that the first integral above is majorized by $\prod(1 + M^2 a_n^2)$ if $|f_n| \leq M$ for all $n$.]

**11.** Prove the following variant of Lemma 3.2. Suppose $\{f_k\}_{k=1}^{\infty}$ is a sequence of identically distributed, mutually independent, $\mathbb{R}^d$-valued functions on a probability space $(X, m)$ each having mean zero and the identity as its covariance matrix. If $s_n = \sum_{k=1}^{n} f_k$, then

$$\limsup_{n\to\infty} m(\{x : \sup_{1\le k\le n} |s_k(x)| > \lambda n^{1/2}\}) = O(\lambda^{-p}), \qquad \text{for } p > 0.$$

[Hint: If $\nu_n$ denotes the distribution measure of $s_n/n^{1/2}$ and $\alpha = \lambda n^{1/2}$, then the right-hand side of (9) equals $\frac{1}{\lambda}\int_{|t|>\lambda} |t|\, d\nu_n(t)$. With $\lambda \ge 1$ fix $M \ge 1$ and write this last integral as the sum of two terms $\lambda^{-1}\int_{|t|>\lambda^M} + \lambda^{-1}\int_{\lambda^M \ge |t|>\lambda}$. Using the fact that $\int \frac{|s_n|^2}{n}\, dm = 1$, the first term is $O(\lambda^{-1-M})$. By the central limit theorem $\lim_{n\to\infty} \lambda^{-1}\int_{\lambda^M\ge|t|>\lambda} |t|\, d\nu_n(t) = O\left(\lambda^{-1}\int_{|t|>\lambda} |t| e^{-|t|^2/2}\, dt\right)$ so the limit of the second term is also $O(\lambda^{-1-M})$.]

**12.** Prove that almost everywhere

$$|B_t(\omega)| = O(t^{1/2+\epsilon}), \qquad \text{as } t \to \infty,$$

for every $\epsilon > 0$. This is the analog of the strong law of large numbers given in Corollary 2.9 of the previous chapter.

[Hint: If $B_T^*(\omega)$ denotes $\sup_{0\le t\le T} |B_t(\omega)|$, then the maximal inequality (14) gives $W(\{B_T^* > \alpha\}) \le \frac{1}{\alpha}\|B_T\|_{L^1} = c'\frac{T^{1/2}}{\alpha}$. If $E_k = \{B_{2^k}^* > 2^{\frac{k}{2}(1+\epsilon)}\}$, then we have $\sum_{k\ge 0} W(E_k) = O\left(\sum_{k\ge 0} 2^{-\frac{k}{2}\epsilon}\right) < \infty$.]

**13.** If $B_t$ is a Brownian motion process then so is ${B'}_t = tB_{1/t}$.

[Hint: Note the continuity of almost all paths of $B'_t$ at the origin follows from the previous exercise. To verify property B-2, use Exercise 29 in the previous chapter.]

**14.** Show that $\limsup_{t\to 0} \frac{|B_t(\omega)|}{t^{1/2}} = \infty$ almost everywhere; hence almost all Brownian paths are not Hölder $1/2$.

Also show that $\limsup_{t\to \infty} \frac{|B_t(\omega)|}{t^{1/2}} = \infty$ almost everywhere; hence almost all Brownian paths exit every ball.

[Hint: By the previous exercise it suffices to check the result when $t \to 0$. Consider $d = 1$. Then

$$W(\{|B_\alpha - B_\beta| > \gamma\}) = \frac{1}{\sqrt{2\pi(\beta-\alpha)}} \int_{|u|>\gamma} e^{-\frac{u^2}{2(\beta-\alpha)}}\, du, \qquad \text{if } \beta > \alpha.$$

Thus

$$W(\{|B_{2^{-k}} - B_{2^{-k+1}}| > 2^{-k/2}\mu_k\}) \geq \frac{1}{\sqrt{2\pi}} \int_{|u|\geq \mu_k} e^{-u^2/2}\, du \geq c_1 e^{-c_2\mu_k^2}.$$

Now choose $\mu_k \to \infty$ so slowly that $\sum_{k\geq 0} e^{-c_2\mu_k^2} = \infty$ and apply the Borel-Cantelli lemma (Exercise 20 in the previous chapter).]

**15.** Calculate the (joint) probability distribution measure of $(B_{t_1}, B_{t_2}, \ldots, B_{t_k})$.
[Hint: Use Exercise 8 (a) in the previous chapter.]

**16.** Show that the following generalization of the fact that $\mathcal{A}_{0+} = \mathcal{A}_0$ holds: if we define $\mathcal{A}_{t+}$ to be $\bigcap_{s>t} \mathcal{A}_s$, then $\mathcal{A}_{t+} = \mathcal{A}_t$.

**17.** The previous exercise gives the right-continuity of the collection $\{\mathcal{A}_s\}$. Prove the following left-continuity for every $t > 0$, $\mathcal{A}_t = \mathcal{A}_{t-}$, where $\mathcal{A}_{t-}$ is the $\sigma$-algebra generated by all $\mathcal{A}_s$ for $s < t$.

[Hint: Consider first cylindrical sets in $\mathcal{A}_t$.]

**18.** Let $\sigma$ be a stopping time. Show that:

(a) $\sigma$ is $\mathcal{A}_\sigma$-measurable.

(b) $B_{\sigma(\omega)}(\omega)$ is $\mathcal{A}_\sigma$-measurable.

(c) $\mathcal{A}_\sigma$ is the $\sigma$-algebra determined by the stopped process $\hat{B}_t$ with $\hat{B}_t(\omega) = B_{t\wedge\sigma(\omega)}(\omega)$.

[Hint: For (a), note that $\{\sigma(\omega) \le \alpha\} \cap \{\sigma(\omega) \le t\} = \{\sigma(\omega) \le \min(\alpha, t)\}$. For (b), show first that for any Borel subset $E$ of $\mathbb{R}^d$ and $t \ge 0$, one has $\{B_{\sigma(\omega)}(\omega) \in E\} \cap \{\sigma \le t\} \in \mathcal{A}_t$ whenever $\sigma$ takes on only discrete values. Then approximate $\sigma$ by $\sigma^{(n)}$ as in the proof of Theorem 5.3.]

**19.** Let $u$ be a bounded Borel measurable function on a bounded open set $\mathcal{R} \subset \mathbb{R}^d$. Suppose that $u$ satisfies the mean-value property on spheres, that is, (21).

(a) Show that if $B$ is a ball contained in $\mathcal{R}$ and centered at $x$, then

$$u(x) = \frac{1}{m(B)} \int_B u(y)\, dy,$$

where $m$ is the Lebesgue measure on $\mathbb{R}^d$.

(b) As a result, the function $u$ is continuous in $\mathcal{R}$ and the argument in Section 4.1, Chapter 5 of Book III shows that the function $u$ is harmonic in $\mathcal{R}$.

[Hint: For (b), show that locally, $u(x) = (u * \varphi)(x)$, where $\varphi$ is a smooth radial function supported on an appropriately small ball and with $\int \varphi = 1$.]

**20.** An bounded open set $\mathcal{R}$ has a **Lipschitz boundary** if $\partial\mathcal{R}$ can be covered by finitely many balls, so that for each such ball $B$, the set $\partial\mathcal{R} \cap B$ can (possibly after a rotation and translation) be written as $x_d = \varphi(x_1, \ldots, x_{d-1})$, where $\varphi$ is a function that satisfies a Lipschitz condition.

Verify that if $\mathcal{R}$ has a Lipschitz boundary, then it satisfies the outside cone condition. Thus, in particular, if $\mathcal{R}$ is of class $C^1$ (in the sense of Section 4 in Chapter 7) then $\mathcal{R}$ satisfies the outside cone condition.

So in these cases the Dirichlet problem is uniquely solvable.

**21.** Suppose $\mathcal{R}_1$ and $\mathcal{R}_2$ are two open and bounded sets in $\mathbb{R}^d$, with $\overline{\mathcal{R}}_1 \subset \mathcal{R}_2$. Let $\mu_1^x$ and $\mu_2^x$ denote the harmonic measures of $\mathcal{R}_1$ and $\mathcal{R}_2$ respectively, as defined at the beginning of Section 5. Show that the following generalization of the mean-value property (21) holds: whenever $x \in \mathcal{R}_1$, then

$$\mu_2^x = \int_{\partial \mathcal{R}_1} \mu_2^y \, d\mu_1^x(y),$$

in the sense that $\mu_2^x(E) = \int_{\partial \mathcal{R}_1} \mu_2^y(E) \, d\mu_1^x(y)$ for any Borel set $E \subset \partial \mathcal{R}_2$.

## Chapter 7. A Glimpse into Several Complex Variables

**1.** Suppose $f$ is holomorphic in a polydisc $\mathbb{P}_r(z^0)$, and assume that $f$ vanishes in a neighborhood of $z^0$. Then $f = 0$ throughout $\mathbb{P}_r(z^0)$.
[Hint: Expand $f(z) = \sum a_\alpha (z - z^0)^\alpha$ in $\mathbb{P}_r(z^0)$, using Proposition 1.1, and note that all $a_\alpha$ are zero.]

**2.** Show that:

(a) If $f$ is holomorphic in a pair $\mathbb{P}_\sigma(z^0)$ and $\mathbb{P}_\tau(z^0)$ of polydiscs centered at $z^0$ with $\sigma = (\sigma_1, \ldots, \sigma_n)$ and $\tau = (\tau_1, \ldots, \tau_n)$, then $f$ extends to be holomorphic in $\mathbb{P}_r(z^0)$, wherever $r = (r_1, \ldots, r_n)$ and $r_j \leq \sigma_j^{1-\theta}\tau_j^\theta$, $1 \leq j \leq n$, for some $0 \leq \theta \leq 1$.

(b) If $S = \{s = (s_1, \ldots, s_n), \quad s_j = \log r_j, \quad \text{where } f \text{ is holomorphic in } \mathbb{P}_r(z^0)\}$, then $S$ is a convex set.

[Hint: Consider $\sum a_\alpha(z - z^0)$ that represents $f$ both in $\mathbb{P}_\sigma(z^0)$ and $\mathbb{P}_\tau(z^0)$.]

**3.** Given $\Omega$ any open subset of $\mathbb{C}^1$, construct a holomorphic function $f$ in $\Omega$ that cannot be continued analytically outside $\Omega$.

[Hint: Given any sequence of points $\{z_j\}$ in $\Omega$, which does not have a limit point in $\Omega$, there exists an analytic function in $\Omega$ vanishing exactly at those $z_j$.]

**4.** Suppose $\Omega$ is a bounded region in $\mathbb{C}^n$, $n > 1$, and $f$ is holomorphic in $\Omega$. Suppose $Z$, the zero set of $f$, is non-empty. Then $\overline{Z}$ intersects $\partial\Omega$, that is, $\overline{Z} \cap \partial\Omega$ is not empty.

[Hint: Let $w$ be a point in $\overline{\Omega}^c$. Let $z^0 \in Z$ be a point furthest from $w$. Define $\gamma$ to be the unit vector in the direction from $z_0$ to $w$, and let $\nu$ be another unit vector so that both $\nu$ and $i\nu$ are perpendicular to $\gamma$. Consider the one-variable function $h_\epsilon(\zeta)$ given by $h_\epsilon(\zeta) = f(z^0 - \epsilon\gamma + \zeta\nu)$. Then for $\epsilon > 0$, the function $h_\epsilon(\zeta)$ does not vanish in a fixed neighborhood of $\zeta = 0$.]

**5.** Suppose $f$ is continuous and has compact support in $\mathbb{C}^1$.

(a) Show that $u = f * \Phi$ in Proposition 3.1 belongs to $\mathrm{Lip}(\alpha)$, for every $\alpha < 1$.

(b) Show that $u$ is not necessarily in $C^1$.

[Hint: For (b) consider $f(z) = z(\log(1/|z|))^\epsilon$ but modified away from the origin to have compact support.]

**6.** Verify the identity in $\mathbb{C}^1$

$$F(z) = \frac{1}{2\pi i}\int_{\partial\Omega}\frac{F(\zeta)}{\zeta - z}\,d\zeta - \frac{1}{\pi}\int_{\Omega}\frac{(\partial F/\partial\overline{\zeta})(\zeta)}{(\zeta - z)}\,dm(\zeta)$$

for appropriate regions $\Omega$ and $C^1$ functions $F$. Use this identity to give an alternative proof of Proposition 3.1.

**7.** Prove the following. The necessary and sufficient condition that the solution $u(z) = \frac{1}{\pi} \int \frac{f(\zeta)}{\zeta - z}\, dm(\zeta)$ of $\partial u/\partial \overline{z} = f$ in $\mathbb{C}^1$, have compact support when $f$ has compact support, is that

$$\int_{\mathbb{C}} \zeta^n f(\zeta)\, dm(\zeta) = 0, \qquad \text{for all } n \geq 0.$$

[Hint: In one direction, note that $\frac{\partial}{\partial \overline{z}}(z^n u(z)) = z^n f(z)$. For the converse, observe that for large $z$, $u(z) = \sum_{n=0}^{\infty} a_n z^{-n-1}$, with $a_n = \frac{1}{\pi} \int \zeta^n f(\zeta)\, dm(\zeta)$.]

**8.** Suppose $\Omega$ is a region in $\mathbb{R}^d$ with a defining function $\rho$ that is of class $C^k$.

(a) If $F$ is a $C^k$ function defined on $\mathbb{R}^d$ and $F = 0$ on $\partial\Omega$, show that $F = a\rho$, with $a \in C^{k-1}$.

(b) Suppose $F_1 = F_2$ on $\partial\Omega$. Show that if $X$ is any tangential vector field then

$$X(F_1)|_{\partial\Omega} = X(F_2)|_{\partial\Omega}.$$

[Hint: Write $F_1 - F_2 = a\rho$.]

**9.** Verify that the extension $F$ given by Theorem 4.1 is the unique solution to the Dirichlet problem for $\Omega$ with boundary data $F_0$.

**10.** Use the region $\{z \in \mathbb{C}^n : \ \rho < |z| < 1\}$ to show that the connectedness hypotheses in Theorem 3.3 and Theorem 4.1 are necessary.

**11.** That the connectedness properties in the hypotheses of Theorems 3.3 and 4.1 are related can be seen as follows. Suppose $\Omega$ is a bounded region with $C^1$ boundary. For $\epsilon > 0$, let $\Omega_\epsilon$ be the "collar" defined by $\{z : \ d(z, \partial\Omega) < \epsilon\}$, and let $\Omega_\epsilon^- = \Omega_\epsilon \cap \Omega$. Then for sufficiently small $\epsilon$ the following are equivalent:

(i) $\overline{\Omega}^c$ is connected,

(ii) $\Omega_\epsilon$ is connected,

(iii) $\Omega_\epsilon^-$ is connected.

[Hint: For instance to see why (ii) or (iii) implies (i), suppose $P_1$ and $P_2$ are two points in $\overline{\Omega}^c$, and let $\Gamma_1$ and $\Gamma_2$ denote the connected components of $\overline{\Omega}^c$ which contain $P_1$ and $P_2$ respectively. Connect $P_1$ to a point $Q_1$ on $\partial\Omega \cap \overline{\Gamma}_1$, and $P_2$ to a point $Q_2$ on $\partial\Omega \cap \overline{\Gamma}_2$. Since $\Omega_\epsilon^-$ is connected one can then connect $Q_1$ to $Q_2$ by a path in $\overline{\Omega}^c$.

Conversely, to show that (i) implies (iii) for example, let $A$ be a point in $\Omega$ and $B$ a point in $\overline{\Omega}^c$. If $P_0$ and $P_1$ belong to $\partial\Omega$, let $\gamma_0$ be any path starting at $A$ traveling in $\Omega$, passing through $P_0$, then traveling in $\Omega^c$ ending at $B$. Similarly, let $\gamma_1$ be path connecting $A$ to $B$ passing through $P_1$. These paths can be constructed because both $\Omega$ and $\overline{\Omega}^c$ are connected. Then, since $\mathbb{C}^n$ is simply-connected, deform the path $\gamma_0$ into $\gamma_1$, and denote such transformation by $s \mapsto \gamma_s$ with $0 \leq s \leq 1$. To conclude, consider the intersection of $\gamma_s$ with $\partial\Omega$.]

**12.** Let $\Omega$ be a simply connected bounded region in $\mathbb{C}^1$ with a boundary of class $C^1$. Suppose $F_0$ is a given continuous function on $\partial\Omega$. Show that a necessary and sufficient condition that there is an $F$, holomorphic in $\Omega$, continuous on $\overline{\Omega}$ so that $F = F_0$ on $\partial\Omega$ is that $\int_{\partial\Omega} z^n F_0(z)\, dz = 0$, for $n = 0, 1, 2, \ldots$ .

[Hint: One direction is clear from Cauchy's theorem. For the converse define $F^{\pm}(z) = \frac{1}{2\pi i}\int_{\partial\Omega} \frac{F_0(\zeta)}{\zeta - z}\, d\zeta$, according to whether $z \in \Omega$ or $z \in \overline{\Omega}^c$. Now the hypothesis implies that $F^+(z) = 0$, $z \in \overline{\Omega}^c$. Also $F^-(z) - F^+(\tilde{z}) \to F_0(\zeta)$, $z \to \zeta$ if $\zeta \in \partial\Omega$, $z \in \Omega$, the segment $[z, \zeta]$ is normal to the tangent line of $\partial\Omega$ at $\zeta$, and $\tilde{z}$ is the reflection of $z$ across that line. That is, $\frac{\tilde{z}+z}{2} = \zeta$, $\tilde{z} \in \overline{\Omega}^c$. The convergence asserted is related to the expression of the delta function given by $i\pi\delta = \frac{1}{2}\left(\frac{1}{x - i0} - \frac{1}{x + i0}\right)$, in Section 2 of Chapter 3.]

**13.** Show that with an additional change of variables, that is, introducing complex coordinates, the canonical representations (16) and (17) of the boundary can be simplified to state

$$y_n = \sum_{j=1}^{n-1} \lambda_j |z_j|^2 + o(|z'|^2), \qquad \text{for } z' \to 0.$$

[Hint: Consider the change of variables $z_n \mapsto z_n - z_n(c_1 z_1 + \cdots + c_{n-1} z_{n-1} + D z_n)$, $z_j \mapsto z_j$, $1 \leq j \leq n-1$, for suitable constants $c_1, \ldots, c_{n-1}$.]

**14.** The fact that when $n = 1$ there are no local holomorphic invariants at boundary points is indicated by the following fact. Suppose $\gamma$ is a $C^k$ curve in $\mathbb{C}^1$. Then for every $z^0 \in \gamma$, there is a holomorphic bijection $\Phi$ of a neighborhood of $z^0$ to a neighborhood of the origin, so that $\Phi(\gamma)$ is the curve $\{y = \varphi(x)\}$, with $\varphi(x) = o(x^k)$ as $x \to 0$.

[Hint: Suppose $y = a_2x^2 + \cdots + a_kx^k + o(x^k)$ as $x \to 0$, and consider $\Phi^{-1}$ defined by $\Phi^{-1}(z) = z + i\left(\sum_{j=2}^k a_j z^j\right)$.]

**15.** Consider the hypersurface $M$ in $\mathbb{C}^3$ given by $M = \{\mathrm{Im}(z_3) = |z_1|^2 - |z_2|^2\}$. Show that $M$ has the remarkable property that any holomorphic function $F$ defined in a neighborhood of $M$ continues analytically into all of $\mathbb{C}^3$.

[Hint: Use Theorem 7.5 to find a fixed ball $B$ centered at the origin so that $F$ continues into all of $B$. Then rescale.]

**16.** That the maximum principle of Theorem 6.1 does not hold in the case $n = 1$ can be seen as follows. Start with $f(e^{i\theta}) \in C^\infty$, so that $f \geq 0$, $f(e^{i\theta}) = 0$ for $|\theta| \leq \pi/2$, $f(e^{i\theta}) = 1$ for $3\pi/4 \leq |\theta| \leq \pi$. Write $f(e^{i\theta}) = \sum_{n=0}^{\infty} a_n e^{in\theta} + \sum_{-\infty}^{n=-1} \overline{a}_n e^{in\theta}$, $G(z) = \sum_{n=0}^{\infty} a_n z^n$, and $F_N(z) = e^{NG(z)}$. Verify that $F_N$ is continuous in the closed disc $|z| \leq 1$, $|F_N(e^{i\theta})| = 1$, for $|\theta| \leq \pi/2$ but $|F_N(z)| \geq c_1 e^{c_2 N(1-|z|)}$ in the closed disc, for two positive constants $c_1$ and $c_2$.

[Hint: $G(z) = u + iv$ where $u(r, \theta) = f * P_r$, with $P_r$ the Poisson kernel.]

**17.** Verify the following:

(a) The inverse of the mapping of $\mathcal{U}$ to the unit ball given in the Appendix is $z_n = i\left(\frac{1-w_n}{1+w_n}\right)$, and $z_k = \frac{w_k}{1+w_n}$, $k = 1, \ldots, n-1$.

(b) For each $(\zeta, t) \in \mathbb{C}^{n-1} \times \mathbb{R}$ consider the following "translation" on $\mathbb{C}^n$, $r_{(\zeta,t)}$ given by

$$r_{(\zeta,t)}(z', z_n) = (z' + \zeta, z_n + t + 2i(z' \cdot \overline{\zeta}) + i|\zeta|^2).$$

Then $r_{(\zeta,t)}$ maps $\mathcal{U}$ and $\partial\mathcal{U}$ to themselves, respectively. Composing these mappings leads to the composition formula

$$(\zeta, t) \cdot (\zeta', t') = (\zeta + \zeta', t + t' + 2\mathrm{Im}(\zeta \cdot \overline{\zeta}')).$$

Under this law $\mathbb{C}^{n-1} \times \mathbb{R}$ becomes the "Heisenberg group."

(c) $\mathcal{U}$ (as well as $\partial\mathcal{U}$) is invariant under the "non-isotropic" dilations $(z', z_n) \to (\delta z', \delta^2 z_n)$, $\delta > 0$.

(d) Both $\mathcal{U}$ and $\partial\mathcal{U}$ are invariant under the mappings $(z', z_n) \mapsto (u(z'), z_n)$, where $u$ is a unitary mapping of $\mathbb{C}^{n-1}$.

**18.** Define $\mathcal{H}_\lambda$ to be the space of functions $f$ holomorphic in $\mathbb{C}^{n-1}$, for which

$$\int_{\mathbb{C}^{n-1}} |f(z)|^2 e^{-4\pi\lambda|z|^2}\, dm(z) = \|f\|^2_{\mathcal{H}_\lambda} < \infty.$$

Show that:

(a) $\mathcal{H}_\lambda$ is trivial if $\lambda \leq 0$.

(b) $\mathcal{H}_\lambda$ is complete in the indicated norm, so $\mathcal{H}_\lambda$ is a Hilbert space.

(c) Define $P_\lambda(f)(z) = \int_{\mathbb{C}^{n-1}} f(w) K_\lambda(z,w) e^{-4\pi\lambda|w|^2}\, dm(w)$, where $K_\lambda(z,w) = (4\lambda)^{n-1} e^{4\pi\lambda z\cdot\overline{w}}$.

Then $P_\lambda$ is the orthogonal projection of $L^2(e^{-4\pi\lambda|w|^2} dm(w))$ to $\mathcal{H}_\lambda$.

[Hint: Show that convergence in the norm $\mathcal{H}_\lambda$ implies uniform convergence on compact subsets of $\mathbb{C}^{n-1}$, using Lemma 8.2.]

**19.** Prove:

(a) The space $\mathcal{H}$ in Section 8.1, is complete, and hence is a Hilbert space.

(b) Show that the Cauchy integral $f \mapsto C(f)$ gives the orthogonal projection from $L^2(\partial\mathcal{U}, d\beta)$ to the linear space of functions $F_0$ that arise as $\lim_{\epsilon\to 0} F_\epsilon$, for $F \in H^2(\mathcal{U})$.

[Hint: For (a), use the previous exercise.]

## Chapter 8. Oscillatory Integrals in Fourier Analysis

**1.** Use spherical coordinates to show that in $\mathbb{R}^d$

$$\int_{S^{d-1}} e^{-2\pi i x\cdot\xi}\, d\sigma = c_d \int_{-1}^{1} e^{-2\pi i|\xi|u}(1-u^2)^{\frac{d-3}{2}}\, du,$$

with $c_d$ the area of the unit sphere $S^{d-2}$ in $\mathbb{R}^{d-1}$. Then deduce formula (3) from Problem 2 in Chapter 6, Book I.

**2.** Let the hypersurface $M$ contain a neighborhood of a hyperplane (for example $\{x_d = 0\}$). Show that in this case $\widehat{d\mu}(\xi) \neq O(|\xi|^{-\epsilon})$, as $|\xi| \to \infty$ for any $\epsilon > 0$.

**3. Principle of stationary phase when $d = 1$.** Consider

$$I(\lambda) = \int_{-\infty}^{\infty} e^{i\lambda\Phi(x)}\psi(x)\,dx,$$

where $\psi$ is a $C^\infty$ function of compact support and $x = 0$ is the only critical point of $\Phi$ in the support of $\psi$, while $\Phi''(0) \neq 0$. Then for every positive integer $N$,

$$I(\lambda) = \frac{e^{i\lambda\Phi(0)}}{\lambda^{1/2}}\left(a_0 + a_1\lambda^{-1} + \cdots + a_N\lambda^{-N}\right) + O(\lambda^{-N-1/2}), \qquad \text{as } \lambda \to \infty.$$

The $a_k$ are determined by $\Phi''(0), \ldots, \Phi^{(2k+2)}(0)$, and $\psi(0), \ldots, \psi^{(2k)}(0)$. In particular $a_0 = \left(\frac{2\pi}{-i\Phi''(0)}\right)^{1/2}\psi(0)$.

Prove this in two steps:

(a) Consider first the special case when $\varphi(x) = x^2$ dealt with by (8).

(b) Pass to the case of general $\varphi$ by a change of variables that brings $\varphi(x)$ to $x^2$ or $-x^2$.

**4.** Suppose $\Phi$ is of class $C^k$ in an interval $[a, b]$ with $k \geq 2$. Assume that $|\Phi^{(k)}(x)| \geq 1$ throughout the interval. Prove the following generalization of Proposition 2.3

$$\left| \int_a^b e^{i\lambda\Phi(x)}\, dx \right| \leq c_k \lambda^{-1/k}.$$

[Hint: Suppose $\Phi^{(k-1)}(x_0) = 0$, and argue by induction as in the proof of Proposition 2.3.]

**5.** Consider the curve $\gamma(t) = (t, t^k)$ in $\mathbb{R}^2$, with $k$ an integer $\geq 2$. Its curvature vanishes nowhere when $k = 2$ and only at the origin, of order $k - 2$, when $k > 2$. Let $d\mu$ be defined by $\int_{\mathbb{R}^2} f\, d\mu = \int_{\mathbb{R}} f(t, t^k)\psi(t)\, dt$, where $\psi$ is a $C^\infty$ function of compact support, and $\psi(0) \neq 0$. Then prove:

(a) $|\widehat{d\mu}(\xi)| = O(|\xi|^{-1/k})$.

(b) However, this decay estimate is optimal, that is, $|\widehat{d\mu}(0, \xi_2)| \geq c|\xi_2|^{-1/k}$ if $\xi_2$ is large.

[Hint: For (a) use Exercise 4. For (b) consider for example the case when $k$ is even and verify that $\int_{-\infty}^{\infty} e^{i\lambda x^k} e^{-x^k}\, dx = c_\lambda(1 - i\lambda)^{-1/k}$.]

**6.** Show that the $(L^p, L^q)$ results for the averaging operator $A$ given by Corollary 4.2 are optimal by proving the following (in, say, the case of the sphere in $\mathbb{R}^3$):

(a) Suppose $f(x)$ vanishes for small $x$ and $f(x) \geq |x|^{-r}$, for $|x| \geq 1$. Then observe $A(f)(x) \geq c|x|^{-r}$ and thus we must always have $q \geq p$. This restriction corresponds to the side of the triangle joining $(0,0)$ and $(1,1)$.

(b) Next let $f = \chi_{B_\delta}$, where $B_\delta$ is the ball of radius $\delta$. Note that if $\delta$ is small $A(\chi_{B_\delta}) \geq c\delta^2$ for $|1 - |x|| < \delta/2$. So $\|f\|_{L^p} \approx \delta^{3/p}$ while $\|A(f)\|_{L^q} \gtrsim \delta^2\delta^{1/q}$. Hence the inequality $\|A(f)\|_{L^q} \leq c\|f\|_{L^p}$ implies $2 + 1/q \geq 3/p$, which corresponds to the side of the triangle joining $(3/4, 1/4)$ and $(1,1)$.

(c) For the third inequality, use duality and (b).

**7.** By refining the argument given in Exercise 6 (b) one can show that the smoothing of degree $(d-1)/2$ asserted in Proposition 1.1 fails when $p \neq 2$.

In the case $p < 2$ and $d = 3$, this can be seen by taking $\delta > 0$ small and setting $f = \varphi_\delta$, where $\varphi_\delta = \varphi(x/\delta)$ and $\varphi$ is a non-negative smooth function of compact support. Here $\|\varphi_\delta\|_{L^p} \approx c\delta^{3/p}$, while $\|\nabla A(\varphi_\delta)\|_{L^p} \gtrsim \delta\delta^{1/p}$. Hence the inequality $\|A(\varphi_\delta)\|_{L^p_1(\mathbb{R}^3)} \leq C\|\varphi_\delta\|_{L^p(\mathbb{R}^3)}$ fails for small delta when $p < 2$.

[Hint: If $c_1 > 0$ is sufficiently small, then $\delta^2 \lesssim A(\varphi_\delta)$ and $|\nabla A(\varphi_\delta)| \gtrsim \delta$, whenever $|1 - |x|| \leq c_1\delta$.]

**8.** Let $M$ be a (local) hypersurface given in coordinates $(x', x_d) \in \mathbb{R}^{d-1} \times \mathbb{R}$ as $\{x_d = \varphi(x')\}$. Suppose $F$ is any continuous function of small support defined in a a neighborhood of $M$ and set $f = F|_M$.

(a) Show that $\lim_{\epsilon \to 0} \frac{1}{2\epsilon} \int_{d(x,M)<\epsilon} F\, dx$ exists and equals $\int_{\mathbb{R}^{d-1}} f(x', \varphi(x'))(1 + |\nabla_{x'} \varphi|^2)^{1/2}\, dx'$. This limit defines the induced Lebesgue measure $d\sigma$ and equals $\int_M f\, d\sigma$.

(b) Suppose $\rho$ is any defining function of $M$. Show that

$$\lim_{\epsilon \to 0} \frac{1}{2\epsilon} \int_{|\rho|<\epsilon} F\, dx = \int_M f \frac{d\sigma}{|\nabla \rho|}.$$

(c) Suppose $h$ is a Schwartz function on $\mathbb{R}$ with $\int_{\mathbb{R}} h(u)\, du = 1$. Then

$$\lim_{\epsilon \to 0} \epsilon^{-1} \int_{\mathbb{R}^d} h(\rho/\epsilon) F\, dx = \int_M f \frac{d\sigma}{|\nabla \rho|}.$$

[Hint: For (c), assume $h$ is even and let $I(t) = \int_{|\rho(x)|<t} F(x)\, dx$. Then

$$\epsilon^{-1} \int h(\rho/\epsilon) F\, dx = \epsilon^{-1} \int_0^\infty h(u/\epsilon) \frac{dI_u}{du}\, du = -\epsilon^{-1} \int_0^\infty (u/\epsilon) h'(u/\epsilon) \left( \frac{1}{u} I_u \right) du.$$

Now use the fact that $-\int_0^\infty u h'(u)\, du = 1/2$ and $\frac{I_u}{2u} \to \int_M f \frac{d\sigma}{|\nabla \rho|}$, as $u \to 0$.]

**9.** Observe the following Euclidean-invariance properties of the principal curvatures of a hypersurface $M$ in $\mathbb{R}^d$. For each $h \in \mathbb{R}^d$ consider the translate $M + h$ of $M$; also for each rotation $r$ of $\mathbb{R}^d$, the rotated surface $r(M)$; and for each $\delta \in \mathbb{R}$, $\delta \neq 0$, the dilated surface $\delta M$. Denote by $\{\lambda_j(x)\}$ the principal curvatures of $M$ at $x$.

(a) Show that $\{\lambda_j(x-h)\}$, $\{\lambda_j(r^{-1}(x))\}$, and $\{\delta^{-2}\lambda_j(x/\delta)\}$ are the principal curvatures of $M+h$, $r(M)$, $\delta M$ at the points $x+h$, $r(x)$ and $\delta x$, respectively.

(b) Consider the cone $\{x_d^2 = |x'|^2,\ x \neq 0\}$ with defining function $\rho = |x'|^2 - x_d^2$. Using (a), show that at $x$ there are $d-2$ principal curvatures that equal $x_d^{-2}$, and one that vanishes.

**10.** Let $f_0(r) = r^{-1/2}(\log r)^{-\delta}$, $0 < \delta < 1$, when $r \geq 2$, and $f_0(r) = 0$ otherwise.

(a) Prove that $\int |J_k(2\pi\rho r)| f_0(r)\, dr = \infty$ for every $\rho > 0$.

(b) Show as a result that, if $p \geq 2d/(d+1)$, then (31) cannot hold for any $q$ when $M$ is the sphere.

**11.** One can prove that the conjectured condition $q \leq \left(\frac{d-1}{d+1}\right) p'$ for $(L^p, L^q)$ restriction cannot hold in a larger range, by the following argument given in the case $d = 2$.

(a) Suppose the inequality (31) holds for some $p$ and $q$. Show that as a result

$$\int_{1-\delta \leq |\xi| \leq 1} |\hat{f}(\xi)|^q \, d\xi \leq c' \delta \|f\|_{L^p}^q \qquad \text{for small } \delta.$$

(b) Next choose $\hat{f}(\xi_1, \xi_2) = \eta((\xi_1 - 1)/\delta)\eta(\xi_2/\delta)$ when $\eta(u) = 1$ if $|u| \geq 1$. That is, $\hat{f}(\xi)$ dominates the characteristic function of a rectangle of approximate side lengths $\delta$ and $\delta^{1/2}$ that fits inside the annulus $1 - \delta \leq |\xi| \leq 1$. Use this to obtain a contradiction $q > \left(\frac{d-1}{d+1}\right) p'$ by letting $\delta \to 0$.

**12.** Connect the operator $e^{it\triangle}$ and the Fourier transform as follows. Let $m_t$ be the multiplication operator $m_t : f(x) \mapsto \frac{1}{(4\pi it)^d} e^{-\frac{i|x|^2}{4t}} f(x)$.

(a) Show that $e^{it\triangle}(f) = i^{-d} m_t (f m_t)^\wedge$ when $t = 1/4\pi$.

(b) Generalize this identity to any $t \neq 0$ by rescaling.

**13.** Let $\mathrm{Ai}(u) = \lim_{N\to\infty} \frac{1}{2\pi} \int_{-N}^{N} e^{i\left(\frac{v^3}{3}+uv\right)}\, dv$.

(a) Show that this limit exists for every $u \in \mathbb{R}$.

(b) Prove that $|\mathrm{Ai}(u)| \leq c(1+|u|)^{-1/4}$.

(c) Moreover, show that $\mathrm{Ai}(u)$ is rapidly decreasing as $u \to \infty$, for $u > 0$.

[Hint: Write $\Phi(r) = \frac{r^3}{3} + ru$, and apply the estimates in Section 2. For (a) use the fact that $\Phi'(r) \to \infty$ as $|r| \to \infty$. For (b), use the fact that $|\Phi'(r)| \geq |u|/2$, when $|r| \leq (\frac{1}{2}|u|)^{1/2}$, while $|\Phi''(r)| \geq 2|r|$ when $|r| > (\frac{1}{2}|u|)^{1/2}$.]

**14.** Suppose $F \in L^2(\mathbb{R}^d \times \mathbb{R})$ and $S(F)(x,t) = i\int_0^t e^{i(t-s)\triangle} F(\cdot,s)\,ds$. Prove that:

(a) For each $t$, $S(F)(\cdot,t) \in L^2(\mathbb{R}^d)$, and

$$\|S(F)(\cdot,t)\|_{L^2(\mathbb{R}^d)} \leq |t|^{1/2}\|F\|_{L^2(\mathbb{R}^d\times\mathbb{R})}.$$

(b) If $F(\cdot,t) = e^{it\triangle}G(\cdot,t)$, then

$$\|G(0,t_1) - G(0,t_2)\|_{L^2(\mathbb{R}^d)} \leq |t_1 - t_2|^{1/2}\|G\|_{L^2(\mathbb{R}^d\times\mathbb{R})}.$$

(c) As a result, $t \mapsto F(0,t)$ is continuous in the $L^2(\mathbb{R}^d)$ norm.

[Hint: For (a) and (b) use the unitarity of $e^{it\triangle}$ and Schwarz's inequality. For (c), approximate $F$ by $C^\infty$ functions of compact support, using (b) and (c).]

**15.** Suppose $u$ is a smooth solution of (54) that decays sufficiently quickly as $|x| \to \infty$. Show that both $\int_{\mathbb{R}^d} |u|^2 \, dx$, and $\int_{\mathbb{R}^d} (\frac{1}{2}|\nabla u|^2 - \frac{\sigma}{\lambda+1}|u|^\lambda) \, dx$ are independent of $t$.

[Hint: For the first, note that $\int_{\mathbb{R}^d} \triangle u \, v \, dx = \int_{\mathbb{R}^d} u \triangle v \, dx$. For the second, observe that $\frac{\partial}{\partial t} \int_{\mathbb{R}^d} |\nabla u|^2 \, dx = -\int_{\mathbb{R}^d} \left( \frac{\partial u}{\partial t} \triangle \overline{u} + \frac{\partial \overline{u}}{\partial t} \triangle u \right) dx$.]

**16.** The following is a converse of Propositions 6.6 and 6.8. Suppose $u(\cdot, t)$ is in $L^2(\mathbb{R}^d)$ for each $t$, with $t \mapsto u(\cdot, t)$ continuous in the $L^2$ norm, and $u(\cdot, 0) = 0$. Assume that $\frac{1}{i}\frac{\partial u}{\partial t} - \triangle u = F$ as distributions, with $F \in L^2(\mathbb{R}^d \times \mathbb{R})$. Then show that $u = S(F)$.

[Hint: Use the following fact. If $H(\cdot, t)$ is in $L^2(\mathbb{R}^d)$ for each $t$, with $t \mapsto H(\cdot, t)$ continuous in the $L^2$ norm, $H(\cdot, 0) = 0$, and $\frac{\partial H}{\partial t} = 0$ in the sense of distributions, then $H = 0$. Apply this to $H(\cdot, t) = e^{-it\triangle}(u(\cdot, t) - S(F)(\cdot, t))$.]

**17.** A solution $u$ of the non-linear Schrödinger equation (54) is uniquely determined by its initial data $f$. Moreover the solution depends continuously on this data. These are two features of the "well-posedness" of the problem and can be stated as follows. Assume $\lambda = \frac{d+4}{d}$ and $q = \frac{2d+4}{d}$.

(a) Suppose $u$ and $v$ are two strong solutions defined for $|t| < a$, having the same initial data $f \in L^2(\mathbb{R}^d)$. Show that $u = v$.

(b) Given $f \in L^2(\mathbb{R}^d)$, prove that there are $\epsilon > 0$ and $a > 0$ (depending on $f$) so that if $\|f - g\|_{L^2} < \epsilon$, and $u$ and $v$ are strong solutions of (54) with initial data $f$ and $g$ respectively, then

$$\|u - v\|_{L^q} \leq c\|f - g\|_{L^2(\mathbb{R}^d)}.$$

Here $L^q = L^q(\mathbb{R}^d \times \{|t| < a\})$.

[Hint: Adapt the argument in Theorem 6.9, and for (a) proceed as follows: note that for small $\ell > 0$

$$\|u\|_{L^q(\mathbb{R}^d \times I)} < \delta \qquad \text{and} \qquad \|v\|_{L^q(\mathbb{R} \times I)} < \delta$$

for all intervals $I$ of length $\leq 2\ell$. Thus

$$\|u - v\|_{L^q} \leq \|\mathcal{M}(u) - \mathcal{M}(v)\|_{L^q} \leq \frac{1}{2}\|u - v\|_{L^q},$$

with $L^q = L^q(\mathbb{R}^d \times \{|t| < \ell\})$, and so $u = v$ for $0 \leq t \leq \ell$. Now use the $t$-translation invariance to apply the same argument for $u(\cdot, t + \ell)$ and $v(\cdot, t + \ell)$, and so on.

For (b) note that by choosing $a$ and $\epsilon$ sufficiently small $\|e^{it\triangle} f\|_{L^q} < \delta/4$ and then $\|e^{it\triangle} g\|_{L^q} < \delta/2$, where $L^q = L^q(\mathbb{R}^d \times \{|t| < a\})$. Now the iteration argument shows that the solutions $u$ and $v$ satisfy $\|u\|_{L^q}, \|v\|_{L^q} < \delta$. Also $\|u - v\|_{L^q} \leq \|S(|u|^{\lambda-1}u - |v|^{\lambda-1}v)\|_{L^q} + c\|f - g\|_{L^2}$. But $\|S(|u|^{\lambda-1}u - |v|^{\lambda-1}v)\|_{L^q} \leq \frac{1}{2}\|u - v\|_{L^q}$, so this proves (b).]

**18.** Consider the Radon transform $\mathcal{R}_B$ defined by

$$\mathcal{R}_B(f)(x', x_d) = \int_{\mathbb{R}^{d-1}} f(y', x_d - B(x', y'))\, dy'$$

$x = (x', x_d) \in \mathbb{R}^{d-1} \times \mathbb{R}$, where $B$ is a fixed non-degenerate bilinear form on $\mathbb{R}^{d-1} \times \mathbb{R}^{d-1}$. We write $B(x', y') = C(x') \cdot y'$, and assume that the dimension $d$ is odd.

Verify that:

(a) $\| \left(\frac{\partial}{\partial x_d}\right)^{\frac{d-1}{2}} \mathcal{R}_B(f)\|^2_{L^2(\mathbb{R}^d)} = c_B \|f\|^2_{L^2}$ for every $f \in \mathcal{S}$, with $c_B = \frac{2(2\pi)^{d-1}}{|\det(C)|}$.

(b) If $(\mathcal{R}_B)^*$ is the (formal) adjoint of $\mathcal{R}_B$, then $\mathcal{R}^*_B = \mathcal{R}_{B^*}$, with $B^*(x, y) = -B(x, y)$. Also $\frac{\partial}{\partial x_d}\mathcal{R}_B = \mathcal{R}_B \frac{\partial}{\partial x_d}$.

(c) Deduce from (a) and (b) the inversion formula

$$\left(i\frac{\partial}{\partial x_d}\right)^{d-1} \mathcal{R}^*_B \mathcal{R}_B(f) = c_B f.$$

**19.** We take the Radon transform $\mathcal{R}_B$ as in the previous exercise (with the dimension $d$ odd) and consider a localized version of it, $\mathcal{R}'_B$, given by

$$\mathcal{R}'_B = \eta' \mathcal{R}_B(\eta f)$$

where $\eta$ and $\eta'$ are a pair of $C^\infty$ functions of compact support. Show that:

(a) $\|\mathcal{R}'_B(f)\|_{L^2} \leq c\|f\|_{L^2}$.

(b) $\left(\frac{\partial}{\partial x}\right)^\alpha \mathcal{R}'_B(f)$ is a finite linear combination of terms of the form $\left(\frac{\partial}{\partial x_d}\right)^\ell (\eta'_\ell \mathcal{R}_B(\eta_\ell(f)))$ with $0 \leq \ell \leq |\alpha|$.

(c) Deduce from the above and part (a) of the previous exercise that $f \mapsto \mathcal{R}'_B(f)$ is a bounded linear transformation from $L^2$ to $L^2_{\frac{d-1}{2}}$.

**20.** The averaging operator from Section 7 satisfies the $L^p$, $L^q$ conclusions stated for the operator $A$ in Corollary 4.2. Prove this by proceeding according to the following steps.

First, recall that $\mathcal{A} = \sum_{k=0}^{\infty} \mathcal{A}_k$, with $\mathcal{A}_k$ given by (65) in Section 7.4, with the sum convergent in the $L^2$ norm. Now fix $r$ and consider

$$T_s = (1 - 2^{1-s})e^{s^2} \sum_{k=0}^{r} 2^{-ks} \mathcal{A}_k.$$

Note that $T_0 = -\sum_{k=0}^{r} \mathcal{A}_k$, and so it will suffice to make $L^p \to L^q$ estimates for $T_0$ that are independent of $r$. Now prove:

(a) $\|T_s(f)\|_{L^2(\mathbb{R}^d)} \leq M\|f\|_{L^2(\mathbb{R}^d)}$ if $\mathrm{Re}(s) = -\frac{d-1}{2}$.

(b) $\|T_s(f)\|_{L^\infty(\mathbb{R}^d)} \leq M\|f\|_{L^1(\mathbb{R}^d)}$ if $\mathrm{Re}(s) = 1$.

Once (a) and (b) have been established, an interpolation via Proposition 4.4 yields

$$\|T_0(f)\|_{L^q} \leq M\|f\|_{L^p},$$

with $p = \frac{d+1}{d}$ and $q = d + 1$, and this leads to the desired conclusion.

[Hint: Part (a) follows from the estimates (70) and (71) for $\alpha = 0$, and the almost-orthogonality argument, Proposition 7.4. To prove (b) note that it suffices to prove that $(1 - 2^{1-s})e^{s^2} \sum_{k=0}^{r} 2^{-ks}\eta(2^{-k}u)$ has a bounded Fourier transform if $\mathrm{Re}(s) = 1$. Let $v$ be the dual variable to $u$. We assume first $|v| \leq 1$. Let $k_0$ be the integer for which $2^{k_0} \leq 1/|v| \leq 2^{k_0+1}$. Now

$$\sum_{k=1}^{r} 2^{-ks} \int \eta(2^{-k}u)e^{2\pi iuv}\, du = \sum_{k \leq k_0} + \sum_{k > k_0}.$$

In the first sum, write $e^{2\pi iuv} = 1 + O(|u||v|)$, and recall that $\eta(\gamma)$ is supported in $1/2 \leq |\gamma| \leq 2$, thus

$$\sum_{k \leq k_0} = O\left(c \sum_{k \leq k_0} 2^{-ks}2^k\right) + O\left(\sum_{k \leq k_0} 2^{-ks} \int \eta(2^{-k}u)\,|v|\;|u|\, du\right),$$

where $c = \int \eta$. However $\sum_{k \leq k_0} 2^{-ks}2^k$ is $O(1/|1 - 2^{1-s}|)$ if $\mathrm{Re}(s) = 1$, while the second term above is (when $\mathrm{Re}(s) = 1$)

$$= O(|v|)\left(\sum_{k \leq k_0} 2^{-k} \int |\eta(2^{-k}u)||u|\, du\right) = O(|v|) \sum_{k \leq k_0} 2^k = O(1).$$

Finally for the second sum, $\sum_{k > k_0}$, integrate by parts, writing $e^{2\pi iuv}$ as $\frac{1}{2\pi iv}\frac{d}{du}(e^{2\pi iuv})$ to obtain a sum that is $O\left(\frac{1}{|v|}\sum_{k > k_0} 2^{-k}\right) = O(2^{-k_0}/|v|) = O(1)$.

If $|v| > 1$, take $k_0 = 0$, and argue similarly.]

**21.** Suppose $\Omega$ is a bounded open convex set with $0 \in \Omega$ and with $C^2$ boundary. Then there is a constant $c > 0$ so that whenever $R \geq 1$ and $\delta \leq 1$, then $x \in R\Omega$ and $|y| \leq \delta$ implies $x + y \in (R + c\delta)\Omega$.

[Hint: One may reduce to the case $R = 1$ by rescaling. To see, for example, that there is a $\mu$ so that $x + y = (1 + \mu\delta)\Omega$ whenever $x \in \partial\Omega$ and $|y| < \delta$ for $\delta$ sufficiently small, proceed as follows. By a Euclidean change of variables, introduce new coordinates so that $x$ has been moved to $(0, 0) \in \mathbb{R}^{d-1} \times \mathbb{R}$, and near that point $\Omega$ is given by $x_d > \varphi(x')$, with $\phi(0) = 0$ and $\nabla_{x'}\varphi(0) = 0$. Then by convexity of $\Omega$, the point corresponding to the initial origin is given by $(z', z_d)$ with $z_d \geq c_1 > 0$. Also $x + y \in (1 + \mu\delta)\Omega$ is equivalent with

$$\frac{y_d + \mu\delta z_d}{1 + \mu\delta} > \varphi\left(\frac{y' + \mu\delta z'}{1 + \mu\delta}\right).$$

Since $|y_d| < \delta$, the left-hand side is $\geq \frac{c_1}{2}\frac{\mu\delta}{1+\mu\delta}$, as soon as $\mu \geq 2/c_1$. Fix such a $\mu$. Now the right-hand side is dominated by

$$A\left|\frac{y' + \mu\delta z'}{1 + \mu\delta}\right|^2 \leq A'\left(\frac{\delta^2 + (\mu\delta)^2}{1 + \mu\delta}\right),$$

and we need only choose $\delta \leq c_2/\mu$, for appropriately small $c_2$.]

**22.** Prove the following two estimates for $r \to \infty$:

(a) $\sum_{1 \le k \le r} d(k)k^\alpha = O\left(r^{\alpha+1} \log r\right)$ if $\alpha > -1$.

(b) $\sum_{r < k} d(k)k^\alpha = O\left(r^{\alpha+1} \log r\right)$ if $\alpha < -1$.

[Hint: Write

$$\sum_{k>r} d(k)k^\alpha = \sum \sum_{mn>r} (mn)^\alpha = \sum_n n^\alpha \left( \sum_{m>r/n} m^\alpha \right)$$
$$= O\left( \sum_n n^\alpha \min(1, (r/n)^{\alpha+1}) \right).$$

**23.** Prove that $rJ_1(r) = \int_0^r \sigma J_0(\sigma)\, d\sigma$, by verifying the following:

(a) $J_1'(r) = \frac{1}{2}\,(J_0(r) - J_2(r))$.

(b) $J_1(r) = \frac{r}{2}\,(J_0(r) + J_2(r))$.

The above shows that $rJ_1'(r) + J_1(r) = rJ_0(r)$, so $\frac{d}{dr}\,(rJ_1(r)) = rJ_0(r)$, proving the assertion.

[Hint: Recall that $J_m(r) = \frac{1}{2\pi}\int_0^{2\pi} e^{ir\sin\theta} e^{-im\theta}\, d\theta$. For (a), differentiate in $r$ under the integral sign. For (b), write $e^{i\theta} = -\frac{1}{i}\frac{d}{d\theta}(e^{-i\theta})$ and integrate by parts.]